Hierarchical and Bidirectional Contrastive Learning for Hyperspectral Image Classification

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***Abstract*— Representing hyperspectral images (HSIs) is a complex and challenging task, primarily due to spectral uncertainty. The learnable prototypical contrastive learning is specialized in discriminative instance representation. However, it requires a much low temperature to prevent model collapse, which can hinder the encoder's ability to capture category relationships. Furthermore, self-supervision at network terminal could obscure specific semantics in hyperfine spectra. In this article, we propose a hierarchical and bidirectional learnable prototypical contrastive learning (HiBiCo) method for HSI representation. We build learnable prototype dictionaries at shallow and final layers of the network for deep contrastive supervision. Moreover, we introduce reverse contrastive learning under negative sample dominance to address excessively uniform representations caused by traditional positive-dominated contrastive loss in dual-dictionary hierarchical supervision. By allowing deep contrastive supervision and two-way information flow along with the general InfoNCE loss, our approach alleviates the uniform distribution, maintains the latent data manifold, and enables diverse and effective representation. Experiments with linear probing demonstrate the effectiveness of our HiBiCo framework in handling complex scenes, highlighting the potential of self-supervised pretraining for HSI representation. The code is available at: https://github.com/sakurashine/HiBiCo**

***Index Terms*— Bidirectional contrastive learning, hierarchical deep supervision, hyperspectral image (HSI), learnable prototypical contrastive learning, self-supervised learning (SSL).**

I. Introduction

HYPERSPECTRAL remote sensing, a powerful combination of imaging and spectroscopy techniques, detects spatial and spectral information of ground objects across a wide range of electromagnetic spectra. It enables object recognition and detection at fine-grained scales, with practical applications in ecological science, urban planning, precision agriculture, and mineral exploration [1], [2], [3], [4]. Hyperspectral image classification (HSIC), which aims to predict labels for each pixel, has been successfully applied in these diverse fields.

High-quality data representation is crucial for ideal image classification. However, this is particularly challenging for hyperspectral images (HSIs), which often contain objects with serious noise interference and spectral uncertainty [5], [6], [7]. While traditional methods focus on feature engineering or advanced classifiers, deep learning conducts feature learning and downstream tasks simultaneously, allowing the encoder to extract crucial features related to the task. Despite many attempts to establish a reliable deep neural network architecture, supervised learning with only a limited amount of manually labeled data often struggles to capture important information from complex scenarios [8], [9], [10]. Instead, it prioritizes task-specific preferences over capturing essential details. As a result, various alternative supervision methods have emerged to relieve time-consuming and labor-intensive data annotation for deep learning. These include active learning [11], metalearning [12], adversarial learning [13], knowledge distillation [14], and other semi-supervised or self-supervised learning (SSL) methods [15], [16], [17].

Large quantities of unlabeled samples provide a more advantageous data-driven hidden pattern recognition compared with a small number of labeled samples. Therefore, SSL presents a more reasonable paradigm for representation learning than supervised methods. It typically pretrains an encoder with a predefined pretext signal. The pretrained encoder is then fine-tuned or directly transferred to a specific downstream task. SSL primarily includes contrastive and generative approaches, with contrastive learning demonstrating more significant potential as it embodies human wisdom. Contrastive SSL trains the encoder to pull positive samples close together and maximize the distance between negatives. This type of learning is effective in discriminating instances and can be beneficial for various remote sensing tasks, such as change detection [18], semantic segmentation [19], and scene classification [20].

Several challenges exist in contrastive learning, such as efficiently sampling positive and negative instances, selecting an appropriate loss function for model optimization, and

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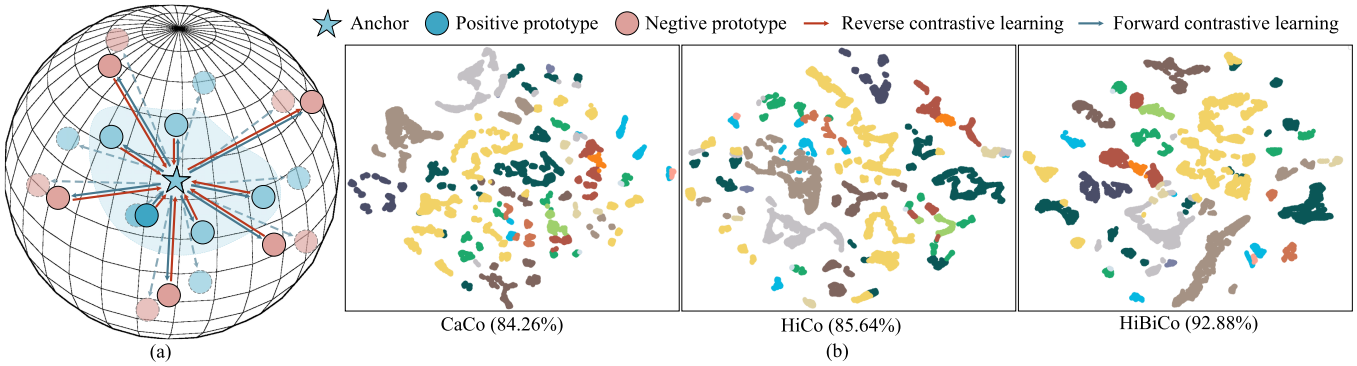


Fig. 1. In learnable prototypical contrastive learning, all prototypes except the most similar one (dark blue cicle) are pushed away from the anchor as negatives (red cicles), as faintly illustrated in (a). The visualized distribution presents worse intraclass clustering, as the results of CaCo [23] are shown in (b). When we perform hierarchical deep contrastive supervision, the situation is improved, as the results achieved by HiCo indicate a 1.38% improvement in accuracy. Furthermore, the HiBiCo integrates HiCo with reverse contrastive learning for bidirectional supervision, by which the potential positives are brought closer to the positive distinguishing boundary, as depicted in (a) with clear solid lines. HiBiCo yields better interclass distinctiveness and intraclass aggregation and demonstrating a noteworthy 7.24% improvement in accuracy.

preventing model collapse. The data augmentation is commonly employed to generate positive pairs, assuming that one instance's semantics remain consistent after augmentation.

The encoder then aims to align the positives semantically and create a uniform distribution for the negatives in a batch or memory bank [21]. Although well-designed data augmentation can improve the encoder's ability to generalize to downstream tasks, instance-level contrastive learning emphasizes learning distinguishable features between instances. This may undermine intraclass consistency, particularly for challenging examples like hard negatives and potential positives. Therefore, learning positive and negative prototypes holds more significant potential, such as adversarial contrast (AdCo) [22] and cooperative-adversarial contrastive learning (CaCo) [23]. However, these methods may lead to a uniform feature distribution when a much low temperature is used to prevent model collapse, especially when dealing with significant interclass similarity and intraclass diversity in HSIs. Thus, sparse prototype sampling and multiple positive learning were considered in [24] to suppress trivial clusters and encourage prototypes that cover more semantic variations.

Current contrastive learning methods have explored the appropriate form of contrast targets by examining positive and negative counterparts. However, little attention has been given to where the contrastive loss should be applied. Typically, contrastive supervision is applied to the final global features, which can result in shallow layers receiving weak knowledge feedback, particularly for layers far from the contrastive loss [25]. When objects in HSI exhibit poor spectral separability, terminal learnable prototypical contrastive learning might overlook subtle spectral differences and disrupt intra-/interclass relationships. DeepMIM [26] verified that intermediate supervision might mislead shallow features in supervised learning but provide better guidance for semantic perception in SSL. Hence, we construct learnable prototype dictionaries at both the shallow and final layers of the network to perform hierarchical contrastive learning (HiCo). The experiment shows that HiCo achieves a 1.38% accuracy improvement compared with the benchmark model under linear probing.

Although HiCo improves HSI representation, a uniformity-tolerance dilemma persists in contrastive learning [27]. In contrastive SSL, the over-uniformity problem typically arises from the over-dispersion of negative samples during model optimization, leading to excessively uniform feature distributions and the collapse of semantic structure [28], [29]. When we set two learnable prototype dictionaries and train positive and negative samples, respectively, each must use a small temperature to prevent model collapse. This approach magnifies detail differences between instances, but it may incorrectly identify potential positive samples as hard negatives, leading to manifold shattered and disrupting the latent semantic relationships. We compared the feature distribution of CaCo [23] and HiCo visualized by t-distributed stochastic neighbor embedding (t-SNE), as shown in Fig. 1(b). We found that despite the improvement in intraclass consistency achieved by HiCo, there are still many trivial clusters. Therefore, adding extra regularization to cluster features is crucial to effectively address over-uniformity in feature learning. Hence, in addition to the general positive-dominated InfoNCE loss used in HiCo, which pulls the positive sample close to the anchor and pushes away all the other negatives, we propose a negative-dominated reverse contrastive loss, as the bidirectional contrastive supervision (BiCo), as illustrated in Fig. 1(a). Besides positive-dominated InfoNCE loss, BiCo further selects the least similar negative sample and pushing the anchor away from it while inversely aggregating the remaining prototypes toward the anchor. The results show that BiCo working with HiCo mitigates overly uniform distribution and achieves an additional 7.24% accuracy.

The main contributions are summarized as follows.

1) We propose a novel deep contrastive self-supervised pretraining method called hierarchical and bidirectional learnable prototypical contrastive learning (HiBiCo) for HSIC. This method improves the discriminative semantic representation of self-supervised algorithms for HSI with high-spectral uncertainty. A series of experiments on four standard hyperspectral datasets demonstrate the superiority of the HiBiCo algorithm.

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2) To enhance the utilization of shallow details in contrastive learning, we introduce learnable prototype dictionaries in both the shallow and final layers of the ResNet backbone for deep contrastive supervision. This improves the model's perceptual ability and more effectively distinguishes objects in HSI with poor spectral separability.
3) We propose a negative-dominated reverse contrastive supervision over the traditional positive-dominated InfoNCE loss to alleviate the over-uniform distribution commonly occurring in learnable prototypical contrastive learning. It enhances the model's semantic representation capability through bidirectional information flow.

The remainder of this article is organized as follows. In Section II, we will introduce the related works about contrastive learning and deep supervision. In Section III, we will detail the proposed HiBiCo method. A comprehensive ablation study and comparative experiments will be presented in Section IV. This article will conclude in Section V.

II. Related Works

A. Contrastive Feature Learning for HSIC

Robust and precise data representation is fundamental for superior image classification; yet, it presents a formidable challenge in HSI data due to pervasive noise interference and inherent spectral variability in the captured scenes. To address these challenges, various approaches have been proposed to enhance feature learning in hyperspectral data. Some studies focus on image unmixing from a spatial or spectral perspective to retrieve discriminative information [5], [30], [31]. Some techniques employ advanced deep networks to effectively capture representative features [32], [33]. These diverse methodologies collectively advance the field of hyperspectral feature learning, addressing the unique challenges.

The contrastive learning represents a significant advancement in discriminative feature learning, with recent works exploring the critical properties of sampling strategy and contrastive loss [34]. The data augmentation is applied to single or multiscale spatial domains of HSI to generate positive samples for contrastive supervision, along with negatives from a batch [35]. To prompt neighborhood consistency, Wang et al. [36] proposed a nearest neighbor-based contrastive learning (NNCNet), where augmented positive instances are replaced with nearest neighbor samples. Data augmentation is challenging in HSI due to its complex spatial distribution and spectral response. Therefore, online training positive and negative samples, such as CaCo [23], is more appropriate. To alleviate over-uniformity in preventing model collapse, Liang et al. [24] proposed a learnable sparse contrastive sampling (LSCoSa) method, which uses learnable sparse prototype sampling to suppress trivial clusters and a multipositive learning strategy to promote features covering more semantic variations.

The InfoNCE loss is often utilized for contrastive learning but usually struggles to balance the alignment and uniformity of features in the hypersphere. This arises from the challenge of distinguishing between potentially positive samples and hard negatives, especially in the context of HSI, where there is significant intraclass diversity and interclass similarity in the spectrum curve. As a result, many efforts have been made to adapt the contrastive loss for HSIC. For instance, Zhao et al. [37] modified a purified InfoNCE loss by introducing super-pixel weighting to mitigate the impact of false negative samples in contrastive learning. Li et al. [38] proposed a similarity-guided group contrastive loss to enhance features with intragroup aggregation and intergroup separation. Ye et al. [39] proposed a two-stream self-supervised framework to align the raw input with its subsets using a symmetric cosine similarity loss without the need for negative samples. Instead of weighting roles of positive and negative examples in contrastive loss, we perform contrastive supervision in a two-way flow to improve the semantic distribution of features.

B. SSL With Deep Supervision

Deep supervision has been studied to assist with training deep networks and potentially mitigate gradient vanishing problems [40], [41]. Zhang et al. [42] highlighted that while task-relevant supervised learning might not be ideal for intermediate-layer features, contrastive learning can offer improved semantic supervision and help prevent overfitting. As a result, they integrated contrastive supervision in the intermediate layers along with task supervision in the final layer, showing that it yielded good performance. DeepMIM [26] further demonstrated that adding extra deep supervision on intermediate layers boosts the shallow features with more semantics and diversities, leading to benefits for self-supervised pretraining. To magnify shallow detailed information essential for fine ground object recognition, Huang et al. [43] developed an HiCo approach with deep supervision on global and local levels for HSIC. In addition, deep supervision has also been used in other fine-grained remote sensing tasks, such as bitemporal change detection [44] and synthetic aperture radar (SAR) ship classification [45].

In our approach, we construct learnable prototype dictionaries at both the shallow and final layers of the network for HiCo. Although it helps to perceive detailed information, additional deep supervision will compound the over-uniformity concomitant with prototypical contrastive learning. Thus, we propose an extra reverse contrastive loss with a negative-dominated approach to capture potential manifolds in twists and turns.

III. Proposed Method

This section will first provide an overview of learnable prototypical contrastive learning for HSIC. Then, we will introduce our proposed HiCo and negative-dominated reverse contrastive loss to enhance contrastive learning for discriminative feature representation.

A. Learnable Prototypical Contrastive Learning for HSIC

To achieve SSL, the learnable prototypical contrastive learning typically involves a Siamese architecture with two encoders: one encoder $\boldsymbol{\phi}_q$ parameterized by $\boldsymbol{\theta}_q$, generates query anchors, and the other $\boldsymbol{\phi}_k$ evolved by exponential

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moving average (EMA) of $\boldsymbol{\theta}_q$ produces the corresponding keys. Simultaneously, positive and negative prototypes are maintained in a learnable dictionary $\mathbf{D}$. The keys retrieve positive examples from $\mathbf{D}$ for anchors, and the contrastive loss is designed to align positives and push away other negatives to uniform all prototypes.

Given a sample $\boldsymbol{x}$ in a mini-batch $\mathcal{B}$, we expect to obtain its discriminative semantic representation z from the online encoder $[z = \boldsymbol{\phi}_q(\boldsymbol{x})]$, which can be transferred to downstream tasks. The key encoder produces key $\hat{z} = \boldsymbol{\phi}_k(\boldsymbol{x})$ for the anchor to retrieve a positive example. Both branches apply data augmentation to the input before feature encoding to assist the Siamese network in asymmetry and generalization. The dictionary $\mathbf{D}$ maintains K learnable prototypes, which are defined as $\mathbf{D} = [\boldsymbol{d}_1, \boldsymbol{d}_2, \ldots, \boldsymbol{d}_K]$. These prototypes are used for sampling positive and negative examples and can be learned directly via cooperative-adversarial InfoNCE loss [23]. Overall, the encoder parameters $\boldsymbol{\theta}_q$ are updated as follows:

$$\boldsymbol{\theta}_q^* = \arg\min_{\boldsymbol{\theta}_q} \sum_{\boldsymbol{x}\in\mathcal{B}} \mathcal{L}_{\mathrm{P}}(\boldsymbol{x}). \tag{1}$$

The contrastive loss $\mathcal{L}_{\mathrm{P}}(\boldsymbol{x})$ is defined as the following prototypical InfoNCE loss [46]:

$$\mathcal{L}_{\mathrm{P}}(\boldsymbol{x}) = -\log \frac{\exp\left(z \cdot \boldsymbol{d}_{j^+}/\tau\right)}{\exp\left(z \cdot \boldsymbol{d}_{j^+}/\tau\right) + \sum_{j\neq j^+} \exp\left(z \cdot \boldsymbol{d}_j/\tau\right)} \tag{2}$$

where the symbol $\cdot$ denotes the inner product of two feature vectors and τ is the temperature parameter controlling the sharpness of the loss. Since the representations z and prototypes $\boldsymbol{d}_j$ are normalized, the inner product yields the cosine similarity between samples. The positive example $\boldsymbol{d}_{j^+}$ is selected from the prototypes in the dictionary that have the maximum probability of similarity with the target representation $\hat{z}$

$$\boldsymbol{d}_{j^+} = \arg\max_{\boldsymbol{d}_j} p\left(\boldsymbol{d}_j \middle| \hat{z}\right) \tag{3}$$

where the probability can be estimated as follows:

$$p\left(\boldsymbol{d}_j \middle| \hat{z}\right)) = \frac{\exp\left(\hat{z} \cdot \boldsymbol{d}_j/\tau\right)}{\sum_{j=1}^{K} \exp\left(\hat{z} \cdot \boldsymbol{d}_j/\tau\right)}. \tag{4}$$

As aforementioned, both positive and negative examples can be directly learned along with the prototypical contrastive loss. This cooperative-adversarial learning mode enables the model to prioritize learning hard negative examples while simultaneously pulling anchors toward their positive prototypes, resulting in more effective contrastive learning. Finally, the overall loss optimization can be defined as follows:

$$\mathbf{D}^*, \boldsymbol{\theta}_q^* = \arg\min_{\boldsymbol{d}_{j^+}, \boldsymbol{\theta}_q} \max_{\boldsymbol{d}_{j\neq j^+}} \sum_{\boldsymbol{x}\in\mathcal{B}} \mathcal{L}_{\mathrm{P}}(\boldsymbol{x}). \tag{5}$$

Contrastive learning with learnable prototypes decouples the model's reliance on large batches. Simultaneously, dictionary prototypes are updated along with network gradients, learning representative key information, capturing highly discriminative features of data, reducing the deliberate design of data augmentation, and improving model generalization. However, contrastive supervision at the network terminal primarily captures image-level relationships, neglecting the variability and complexity of intermediate-layer features. Deep supervision has proven its effectiveness in SSL by adding additional supervision at the middle layer. Therefore, we propose a hierarchical prototypical contrastive learning method.

B. Hierarchical Prototypical Contrastive Learning

Shallow layers in deep models mainly capture local details while deep layers learn global, high-level semantics. HSI detail perception can easily be plagued in Contrastive SSL with terminal supervision. On this basis, we propose a hierarchical prototypical contrast to learn a global discriminative representation and simultaneously capture the spectral differences preserved more in shallow layers, ignored by terminal supervision. As illustrated in Fig. 2, we establish learnable prototype dictionaries at both the shallow layer and the network terminal, enabling local and global HiCo.

Specifically, two dictionaries $\mathbf{D}_s \triangleq \{\boldsymbol{d}_j^{(s)} | j = 1, \ldots, K_s\}$ and $\mathbf{D}_t \triangleq \{\boldsymbol{d}_j^{(t)} | j = 1, \ldots, K_t\}$ are constructed, respectively, behind the shallow layer and terminal of the encoder, providing two groups of positive and negative prototypes at different scales. Therefore, for any input $\boldsymbol{x}$, the positive prototypes can be sampled separately as follows:

$$\boldsymbol{d}_{j^+}^{(s)} = \arg\max_{\boldsymbol{d}_j^{(s)}} p\left(\boldsymbol{d}_j^{(s)} \middle| \hat{z}^{(s)}\right), \quad j = 1, 2, \ldots, K_s \tag{6}$$

$$\boldsymbol{d}_{j^+}^{(t)} = \arg\max_{\boldsymbol{d}_j^{(t)}} p\left(\boldsymbol{d}_j^{(t)} \middle| \hat{z}^{(t)}\right), \quad j = 1, 2, \ldots, K_t. \tag{7}$$

Here, $\hat{z}^{(s)} = \boldsymbol{\phi}_k^s(\boldsymbol{x})$ and $\hat{z}^{(t)} = \boldsymbol{\phi}_k^t(\boldsymbol{x})$ represent the key embeddings output from shallow layer $\boldsymbol{\phi}_k^s(\cdot)$ and terminal layer $\boldsymbol{\phi}_k^t(\cdot)$, respectively. The positive examples are retrieved according to similarity calculation of $\hat{z}^{(i)}$ and prototypes $\boldsymbol{d}_j^{(i)}$, where $i \in \{s, t\}$, as in (4). Experiments have shown that optimal performance is achieved when the shallow dictionary is constructed after the first convolutional layer.

As part of the end-to-end training network, the dictionaries $\mathbf{D}_s$ and $\mathbf{D}_t$ are both optimized through a cooperative-adversarial manner with the contrastive loss similar to (5)

$$\mathbf{D}_s^*, \mathbf{D}_t^*, \boldsymbol{\theta}_q^* = \arg\min_{\boldsymbol{d}_{j^+}^{(s,t)}, \boldsymbol{\theta}_q} \max_{\boldsymbol{d}_{j\neq j^+}^{(s,t)}} \sum_{\boldsymbol{x}\in\mathcal{B}} \mathcal{L}_{\mathrm{HiCo}}(\boldsymbol{x}) \tag{8}$$

where the hierarchical contrastive loss $\mathcal{L}_{\mathrm{HiCo}}(\boldsymbol{x})$ includes the prototypical InfoNCE loss at two dictionaries

$$\mathcal{L}_{\mathrm{HiCo}}(\boldsymbol{x}) = \mathcal{L}_{\mathrm{P}}^{(s)}(\boldsymbol{x}) + \mathcal{L}_{\mathrm{P}}^{(t)}(\boldsymbol{x}) \tag{9}$$

where

$$\mathcal{L}_{\mathrm{P}}^{(i)}(\boldsymbol{x}) = -\log \frac{\exp\left(z^{(i)} \cdot \boldsymbol{d}_{j^+}^{(i)}/\tau\right)}{\exp\left(z^{(i)} \cdot \boldsymbol{d}_{j^+}^{(i)}/\tau\right) + \sum_{j\neq j^+} \exp\left(z^{(i)} \cdot \boldsymbol{d}_j^{(i)}/\tau\right)}. \tag{10}$$

The hierarchical anchors $z^{(s)} = \boldsymbol{\phi}_q^s(\boldsymbol{x})$ and $z^{(t)} = \boldsymbol{\phi}_q^t(\boldsymbol{x})$ of the input align with positive prototypes $\boldsymbol{d}_{j^+}^{(s)}$ and $\boldsymbol{d}_{j^+}^{(t)}$ chosen from dictionaries $\mathbf{D}_s$ and $\mathbf{D}_t$, respectively.

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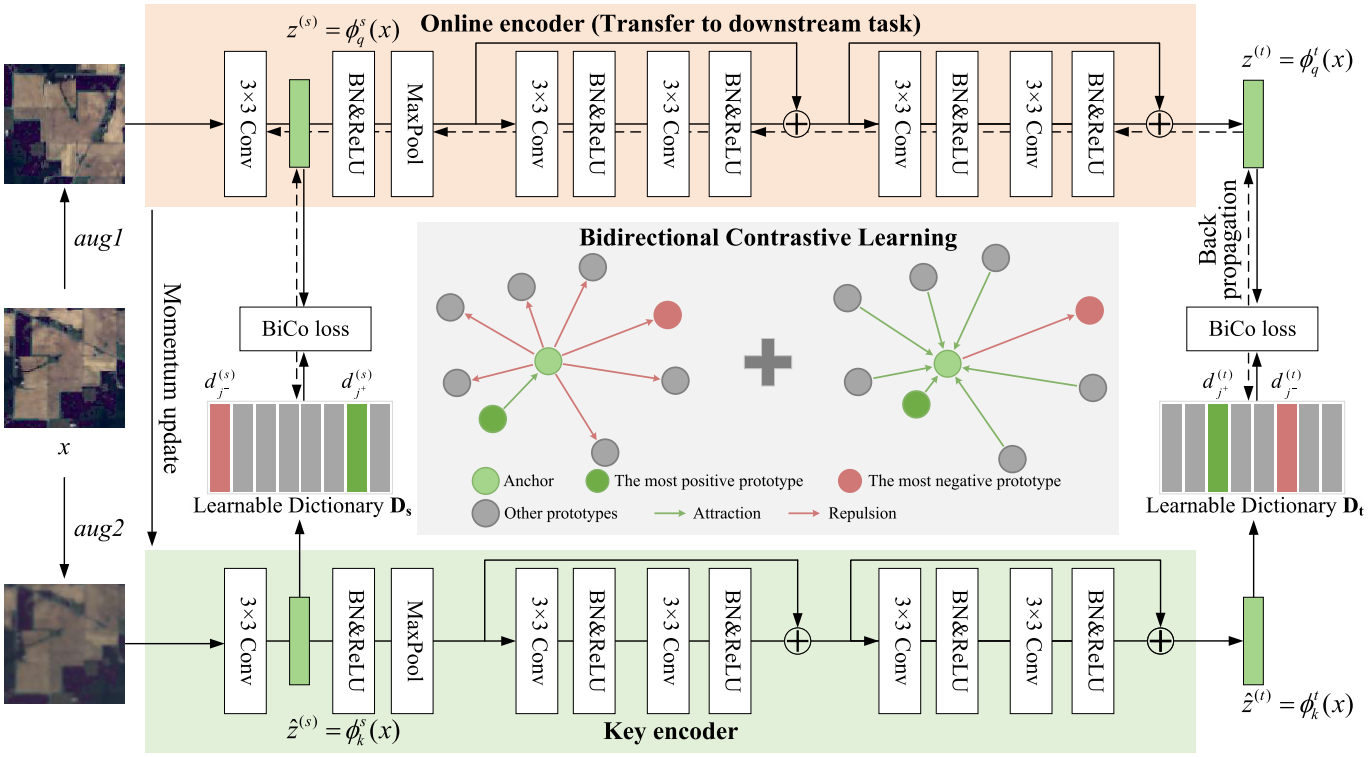


Fig. 2. Overview of our proposed HiBiCo framework. HiBiCo employs a Siamese architecture, one encoder for online feature learning and a key encoder for target consistency. We maintain dictionaries with learnable prototypes at the early layer and the network terminal for hierarchical anchor querying. The most positive and the most negative prototypes are dynamically sampled through cosine similarity calculation between dictionary prototypes and keys. Guided with the sampled prototype, InfoNCE and the negative-dominated reverse contrastive loss are exerted bidirectionally, denoted by BiCo, for model pretraining.

Through deep supervision, HiCo can enhance the encoder's ability to perceive shallow semantics in images and improve its capacity to transfer knowledge to downstream tasks. However, the uniformity-tolerance dilemma [27] remains unresolved. Worse still, it may degrade the underlying manifold clusters of HSI when preventing the model from collapsing because shallow features are more vulnerable to noise. Consequently, we further propose a negative-dominated reverse contrastive loss to improve the distribution of contrast prototypes and alleviate over-uniformity.

C. Bidirectional Prototypical Contrastive Learning

1) Forward Contrast (Standard InfoNCE): Let $\boldsymbol{x}$ be an input sample from a batch, and $\boldsymbol{z}^{(i)} = \boldsymbol{\phi}_q^i(\boldsymbol{x})$ be its latent embedding generated from the shallow layer ($i = s$) or the terminal layer ($i = t$) of the online encoder. In prototypical contrastive learning, we maintain a dynamic dictionary $\mathbf{D}_i = \{\boldsymbol{d}_1^{(i)}, \boldsymbol{d}_2^{(i)}, \ldots, \boldsymbol{d}_K^{(i)}\}$ containing prototype embeddings that cooperatively–adversarially learned via InfoNCE loss. The goal is to learn discriminative features by contrasting $\boldsymbol{z}^{(i)}$ against prototypes in $\mathbf{D}_i$.

The traditional contrastive loss aligns the anchor $\boldsymbol{z}^{(i)}$ with its most confident positive prototype $\boldsymbol{d}_{j^+}^{(i)}$ while repelling all other prototypes as negatives, as shown in (10). $\boldsymbol{d}_{j^+}^{(i)}$ is selected as follows:

$$\boldsymbol{d}_{j^+}^{(i)} = \arg\max_{\boldsymbol{d}_j^{(i)}} p\left(\boldsymbol{d}_j^{(i)} \middle| \hat{\boldsymbol{z}}^{(i)}\right), \quad i \in \{s, t\} \tag{11}$$

where

$$p\left(\boldsymbol{d}_j^{(i)} \middle| \hat{\boldsymbol{z}}^{(i)}\right) = \frac{\exp\left(\hat{\boldsymbol{z}}^{(i)} \cdot \boldsymbol{d}_j^{(i)}/\tau\right)}{\sum_{j=1}^{K} \exp\left(\hat{\boldsymbol{z}}^{(i)} \cdot \boldsymbol{d}_j^{(i)}/\tau\right)}. \tag{12}$$

The phenomenon of over-uniformity often occurs because potential positive examples are mistakenly considered as negatives and treated equally with the actual negative examples in contrastive learning. This means we sacrifice potential manifolds in exchange for avoiding model collapse. Therefore, multiple positive sampling or distribution regularization is applied to help it balance intercluster separability and intracluster diversity [24], [47], [48]. In this article, we propose a novel approach, known as reverse contrastive learning, to alleviate excessive uniform distribution.

2) Reverse Contrastive Learning: To mitigate over-uniformity, we introduce a reverse contrastive loss $\mathcal{L}_{\mathrm{N}}^{(i)}$, which explicitly repels the most confident negative prototype $\boldsymbol{d}_{j^-}^{(i)}$ while attracting other prototypes

$$\mathcal{L}_{\mathrm{N}}^{(i)}(\boldsymbol{x}) = \log \frac{\exp\left(\boldsymbol{z}^{(i)} \cdot \boldsymbol{d}_{j^-}^{(i)}/\tau\right)}{\exp\left(\boldsymbol{z}^{(i)} \cdot \boldsymbol{d}_{j^-}^{(i)}/\tau\right) + \sum_{j \neq j^-} \exp\left(\boldsymbol{z}^{(i)} \cdot \boldsymbol{d}_j^{(i)}/\tau\right)} \tag{13}$$

where $\boldsymbol{d}_{j^-}^{(i)}$ corresponds to the prototype least similar to $\hat{\boldsymbol{z}}^{(i)}$

$$\boldsymbol{d}_{j^-}^{(i)} = \arg\min_{\boldsymbol{d}_j^{(i)}} p\left(\boldsymbol{d}_j^{(i)} \middle| \hat{\boldsymbol{z}}^{(i)}\right). \tag{14}$$

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The loss $\mathcal{L}_{\mathrm{N}}^{(i)}$ shares the same numerator–denominator structure as InfoNCE but removes the negative sign and swaps the roles of the most positive/negative prototypes. The gradient behavior of $\mathcal{L}_{\mathrm{N}}^{(i)}$ is as follows.

1) *For $\boldsymbol{d}_{j^-}^{(i)}$:* Minimizes $z^{(i)} \cdot \boldsymbol{d}_{j^-}^{(i)}$ (repulsion).
2) *For other prototypes $\boldsymbol{d}_{j}^{(i)}$ $(j \neq j^-)$:* Maximizes $z^{(i)} \cdot \boldsymbol{d}_{j}^{(i)}$ (attraction).

Taking the most confident negative examples as a reference, the reverse information flow provides enhanced opportunities for intraclass information aggregation. Reverse contrastive learning effectively delineates the boundary and preserves feature distribution through positive attraction.

3) Bidirectional Contrastive Loss: The bidirectional contrastive loss $\mathcal{L}_{\mathrm{BiCo}}^{(i)}$ combines both forward and reverse contrastive objectives

$$\begin{aligned}\mathcal{L}_{\mathrm{BiCo}}^{(i)}(\boldsymbol{x}) &= \mathcal{L}_{\mathrm{P}}^{(i)}(\boldsymbol{x}) + \lambda \mathcal{L}_{\mathrm{N}}^{(i)}(\boldsymbol{x}) \\ &= -\log \frac{\exp\left(z^{(i)} \cdot \boldsymbol{d}_{j^+}^{(i)}/\tau\right)}{\sum_{j=1}^{K} \exp\left(z^{(i)} \cdot \boldsymbol{d}_{j}^{(i)}/\tau\right)} \\ &\quad + \lambda \log \frac{\exp\left(z^{(i)} \cdot \boldsymbol{d}_{j^-}^{(i)}/\tau\right)}{\sum_{j=1}^{K} \exp\left(z^{(i)} \cdot \boldsymbol{d}_{j}^{(i)}/\tau\right)}\end{aligned} \tag{15}$$

where $\lambda \in [0, 1]$ balances the two terms.

Let us examine the derivatives of positive and negative prototypical similarities under both standard InfoNCE and the proposed bidirectional contrastive loss. It is not hard to show that the gradients of the standard InfoNCE loss with respect to the most confident positive similarity and the most confident negative similarity are formulated as follows:

$$\begin{cases} \dfrac{\partial \mathcal{L}_{P}^{(i)}(\boldsymbol{x})}{\partial\left(z^{(i)} \cdot \boldsymbol{d}_{j^+}^{(i)}\right)} = \dfrac{1}{\tau}\left[-1 + p\left(\boldsymbol{d}_{j^+}^{(i)} | z^{(i)}\right)\right] \\ \dfrac{\partial \mathcal{L}_{P}^{(i)}(\boldsymbol{x})}{\partial\left(z^{(i)} \cdot \boldsymbol{d}_{j^-}^{(i)}\right)} = \dfrac{1}{\tau} p\left(\boldsymbol{d}_{j^-}^{(i)} | z^{(i)}\right). \end{cases} \tag{16}$$

In contrast, the derivative of the proposed bidirectional contrastive loss in updating them is

$$\begin{cases} \dfrac{\partial \mathcal{L}_{\mathrm{BiCo}}^{(i)}(\boldsymbol{x})}{\partial\left(z^{(i)} \cdot \boldsymbol{d}_{j^+}^{(i)}\right)} = \dfrac{1}{\tau}\left[-1 + p\left(\boldsymbol{d}_{j^+}^{(i)} | z^{(i)}\right)\right] \\ \qquad\qquad + \underbrace{\left[-\dfrac{\lambda}{\tau} p\left(\boldsymbol{d}_{j^+}^{(i)} | z^{(i)}\right)\right]}_{\text{always negative}} \\ \dfrac{\partial \mathcal{L}_{\mathrm{BiCo}}^{(i)}(\boldsymbol{x})}{\partial\left(z^{(i)} \cdot \boldsymbol{d}_{j^-}^{(i)}\right)} = \dfrac{1}{\tau} p\left(\boldsymbol{d}_{j^-}^{(i)} | z^{(i)}\right) + \underbrace{\dfrac{\lambda}{\tau}\left[1 - p\left(\boldsymbol{d}_{j^-}^{(i)} | z^{(i)}\right)\right]}_{\text{always positive}}. \end{cases} \tag{17}$$

Compared with standard InfoNCE, $\mathcal{L}_{\mathrm{BiCo}}^{(i)}$ introduces the following.

1) An extra negative term in the positive similarity gradient, enhancing attraction on the most confident positive;
2) An extra positive term in the negative similarity gradient, explicitly promoting separation from the most confident negatives while implicitly maintaining attraction to the remaining samples.

This bidirectional mechanism allows the model to preserve local manifolds by attracting ambiguous prototypes (potential positives) through $\mathcal{L}_{\mathrm{N}}$. Meanwhile, it enhances global separation by repelling the hardest negative prototype via $\mathcal{L}_{\mathrm{N}}$ and attracting the hardest positive via $\mathcal{L}_{\mathrm{P}}$.

4) Bidirectional Contrastive Learning on Prototype Updates: In cooperative-adversarial prototypical contrastive learning mode [23], the positive prototype $\boldsymbol{d}_{j^+}^{(i)}$ and other negative counterparts $\boldsymbol{d}_{j}^{(i)}$ $(j \neq j^+)$ are updated by minimizing and maximizing (2)

$$\begin{cases} \dfrac{\partial \mathcal{L}_{\mathrm{P}}^{(i)}(\boldsymbol{x})}{\partial \boldsymbol{d}_{j^+}^{(i)}} = -\dfrac{1}{\tau}\left[1 - p\left(\boldsymbol{d}_{j^+}^{(i)} | z^{(i)}\right)\right] z^{(i)} \\ \dfrac{\partial \mathcal{L}_{\mathrm{P}}^{(i)}(\boldsymbol{x})}{\partial \boldsymbol{d}_{j}^{(i)}} = \dfrac{1}{\tau} p\left(\boldsymbol{d}_{j}^{(i)} | z^{(i)}\right) z^{(i)}, \quad j \neq j^+. \end{cases} \tag{18}$$

Then, they are updated as follows:

$$\begin{cases} \boldsymbol{d}_{j^+}^{(i)} \leftarrow \boldsymbol{d}_{j^+}^{(i)} + \dfrac{\eta}{\tau}\left[1 - p\left(\boldsymbol{d}_{j^+}^{(i)} | z^{(i)}\right)\right] z^{(i)} \\ \boldsymbol{d}_{j}^{(i)} \leftarrow \boldsymbol{d}_{j}^{(i)} + \dfrac{\eta}{\tau} p\left(\boldsymbol{d}_{j}^{(i)} | z^{(i)}\right) z^{(i)}, \quad j \neq j^+ \end{cases} \tag{19}$$

where η is the learning rate for the gradient update.

In contrast, the derivative of the proposed bidirectional loss in updating a positive prototype $\boldsymbol{d}_{j^+}^{(i)}$ is

$$\begin{aligned}\frac{\partial \mathcal{L}_{\mathrm{BiCo}}^{(i)}(\boldsymbol{x})}{\partial \boldsymbol{d}_{j^+}^{(i)}} &= -\frac{z^{(i)}}{\tau} + (1-\lambda)\frac{z^{(i)}}{\tau}\frac{\exp\left(z^{(i)} \cdot \boldsymbol{d}_{j^+}^{(i)}/\tau\right)}{\sum_{j=1}^{K} \exp\left(z^{(i)} \cdot \boldsymbol{d}_{j}^{(i)}/\tau\right)} \\ &= -\frac{z^{(i)}}{\tau}\left[1 - (1-\lambda) p\left(\boldsymbol{d}_{j^+}^{(i)} | z^{(i)}\right)\right].\end{aligned} \tag{20}$$

Likewise, the gradient of the composite loss $\mathcal{L}_{\mathrm{BiCo}}^{(i)}$ with respect to prototypes $\boldsymbol{d}_{j}^{(i)}$ can be derived as follows:

$$\begin{cases} \dfrac{\partial \mathcal{L}_{\mathrm{BiCo}}^{(i)}(\boldsymbol{x})}{\partial \boldsymbol{d}_{j^+}^{(i)}} = -\dfrac{1}{\tau}\left[1 - (1-\lambda) p\left(\boldsymbol{d}_{j^+}^{(i)} | z^{(i)}\right)\right] z^{(i)} \\ \dfrac{\partial \mathcal{L}_{\mathrm{BiCo}}^{(i)}(\boldsymbol{x})}{\partial \boldsymbol{d}_{j^-}^{(i)}} = \dfrac{1}{\tau}\left[\lambda + (1-\lambda) p\left(\boldsymbol{d}_{j^-}^{(i)} | z^{(i)}\right)\right] z^{(i)} \\ \dfrac{\partial \mathcal{L}_{\mathrm{BiCo}}^{(i)}(\boldsymbol{x})}{\partial \boldsymbol{d}_{j}^{(i)}} = \dfrac{1}{\tau}(1-\lambda) p\left(\boldsymbol{d}_{j}^{(i)} | z^{(i)}\right) z^{(i)} \quad j \notin \{j^+, j^-\}. \end{cases} \tag{21}$$

Then, they are updated as follows:

$$\begin{cases} \boldsymbol{d}_{j^+}^{(i)} \leftarrow \boldsymbol{d}_{j^+}^{(i)} + \dfrac{\eta}{\tau}\left[1 - (1-\lambda) p\left(\boldsymbol{d}_{j^+}^{(i)} | z^{(i)}\right)\right] z^{(i)} \\ \boldsymbol{d}_{j^-}^{(i)} \leftarrow \boldsymbol{d}_{j^-}^{(i)} + \dfrac{\eta}{\tau}\left[\lambda + (1-\lambda) p\left(\boldsymbol{d}_{j^-}^{(i)} | z^{(i)}\right)\right] z^{(i)} \\ \boldsymbol{d}_{j}^{(i)} \leftarrow \boldsymbol{d}_{j}^{(i)} + (1-\lambda)\dfrac{\eta}{\tau} p\left(\boldsymbol{d}_{j}^{(i)} | z^{(i)}\right) z^{(i)}, \quad j \notin \{j^+, j^-\}. \end{cases} \tag{22}$$

From this update, we can observe that the following hold.

1) For the most confident positive prototype, the attraction force toward anchors is strengthened by introducing a coefficient $1 - \lambda < 1$ to their sampling probability. This modification enhances positive sampling certainty.

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Algorithm 1 HiBiCo

Inputs:
online encoder $\boldsymbol{\phi}_q$ and initialized parameter $\boldsymbol{\theta}_q$
key encoder $\boldsymbol{\phi}_k$ and initialized parameter $\boldsymbol{\theta}_k$
initialized dictionary $\mathbf{D}_s$,$\mathbf{D}_t$ of size K_s,K_t
Batch size N, momentum m, temperature τ
for sampled mini batch $\mathcal{B}$ **do**
for all $\boldsymbol{x}$ in batch $\mathcal{B}$ **do**

$$\boldsymbol{z}^{(s)}, \boldsymbol{z}^{(t)} = \boldsymbol{\phi}_q^s(aug1(\boldsymbol{x})), \boldsymbol{\phi}_q^t(aug1(\boldsymbol{x}))$$
$$\hat{\boldsymbol{z}}^{(s)}, \hat{\boldsymbol{z}}^{(t)} = \boldsymbol{\phi}_k^s(aug2(\boldsymbol{x})), \boldsymbol{\phi}_k^t(aug2(\boldsymbol{x}))$$
$$\boldsymbol{d}_{j^+}^{(s)} = \arg\max_{\boldsymbol{d}_j^{(s)}} p(\boldsymbol{d}_j^{(s)}|\hat{\boldsymbol{z}}^{(s)}), \quad j = 1, 2, \ldots, K_s$$
$$\boldsymbol{d}_{j^-}^{(s)} = \arg\min_{\boldsymbol{d}_j^{(s)}} p(\boldsymbol{d}_j^{(s)}|\hat{\boldsymbol{z}}^{(s)}), \quad j = 1, 2, \ldots, K_s$$
$$\boldsymbol{d}_{j^+}^{(t)} = \arg\max_{\boldsymbol{d}_j^{(t)}} p(\boldsymbol{d}_j^{(t)}|\hat{\boldsymbol{z}}^{(t)}), \quad j = 1, 2, \ldots, K_t$$
$$\boldsymbol{d}_{j^-}^{(t)} = \arg\min_{\boldsymbol{d}_j^{(t)}} p(\boldsymbol{d}_j^{(t)}|\hat{\boldsymbol{z}}^{(t)}), \quad j = 1, 2, \ldots, K_t$$

end for
$\mathcal{L} = \sum_{\boldsymbol{x}\in\mathcal{B}} \sum_{i\in\{s,t\}} \mathcal{L}_{\text{BiCo}}^{(i)}(\boldsymbol{x})$
update $\boldsymbol{\phi}_q$, $\mathbf{D}_s$ and $\mathbf{D}_t$ according to Eq. (23)
$\boldsymbol{\theta}_k = m\boldsymbol{\theta}_k + (1-m)\boldsymbol{\theta}_q$
end for
Output: encoder $\boldsymbol{\phi}_q$

2) For other negatives, the attraction force is attenuated (scaled by $1-\lambda$ instead of 1), which prevents excessive concentration around anchors to mitigate generation of hard negatives, and maintains feature diversity by avoiding over-uniform distributions.
3) For the most confident negative prototype, it receives stronger attraction (boosted by λ) compared with other negatives, effectively preventing "useless negative" and ensuring meaningful gradient contributions from all negatives.

D. Overall Loss Function

Finally, the overall loss function of HiBiCo could be expressed as follows:

$$\arg\min_{\theta_q, \boldsymbol{d}_j^{(s,t)}} \sum_{\boldsymbol{x}\in\mathcal{B}} \sum_{i\in\{s,t\}} \mathcal{L}_{\text{BiCo}}^{(i)}(\boldsymbol{x})$$
$$\text{s.t.} \quad \mathbf{D}_s^*, \mathbf{D}_t^* = \sum_{\boldsymbol{x}\in\mathcal{B}} \left(\min_{\boldsymbol{d}_{j^+}^{(s,t)}} \max_{\boldsymbol{d}_{j\neq j^+}^{(s,t)}} \mathcal{L}_{\text{HiCo}}(\boldsymbol{x}) \right). \tag{23}$$

Under bidirectional contrastive learning hierarchically, the features extracted by the encoder tend to exhibit more intra-class clustering and interclass separability in their distribution. This target leads to a substantial improvement in the model's discriminability and generalization. Algorithm 1 summarizes our proposed method HiBiCo.

IV. Experiments and Analysis

In this section, a comprehensive set of experiments is conducted to validate the effectiveness of our proposed HiBiCo, including ablation experiments, hyperparameter analysis, and performance comparison with several state-of-the-art HSI representation and classification models.

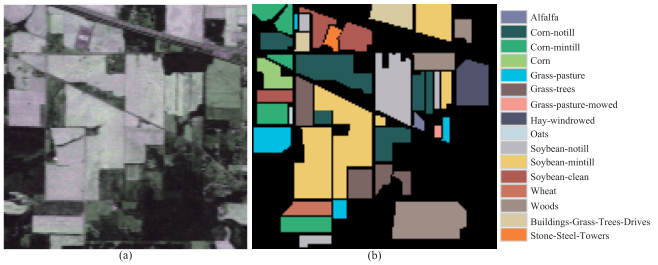


Fig. 3. IP dataset. (a) False-color image. (b) Ground truth.

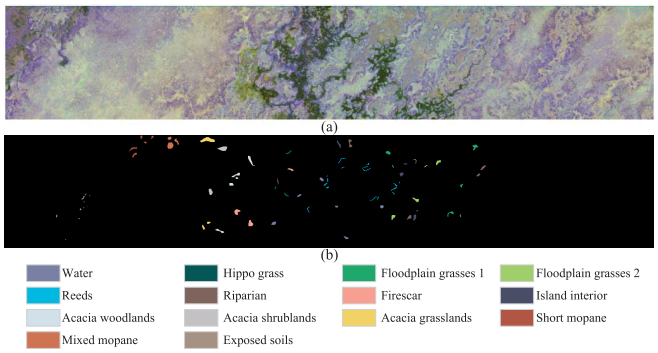


Fig. 4. BO dataset. (a) False-color image. (b) Ground truth.

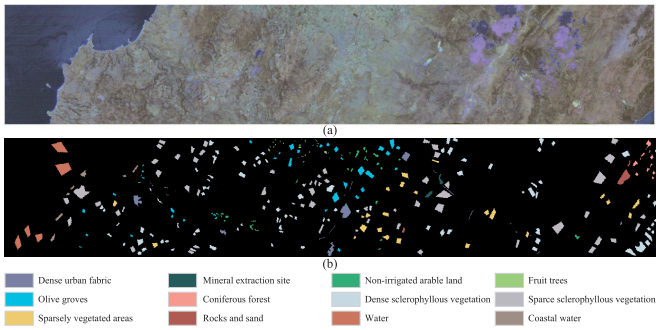


Fig. 5. HR-D dataset. (a) False-color image. (b) Ground truth.

A. Experimental Datasets

We conduct experiments using four benchmark datasets: Indian Pines (IP), Botswana (BO), HyRANK-Dioni (HR-D), and Houston University 2018 (HU18).

1) Indian Pines: The IP dataset was acquired by the AVIRIS sensor over northwest Indiana, consisting of 145 × 145 pixels and 200 spectral bands. It includes ground truth data for 16 land cover classes. Fig. 3(a) and (b) illustrates its false-color image and ground-truth map, respectively.

2) Botswana: The BO dataset was obtained from the NASA EO-1 satellite over the Okavango Delta in BO. It comprises 1476 × 256 pixels and features 145 spectral bands, with ground-truth data for 14 classes. Fig. 4(a) and (b) presents its false-color image and ground-truth map, respectively.

3) HyRANK-Dioni: The HR-D dataset, one of the five HSIs captured by the EO-1 Hyperion sensor, consists of 250 × 1376 pixels and 176 spectral bands. The ground-truth data include 12 classes. Fig. 5(a) and (b) shows its false-color image and ground-truth map, respectively.

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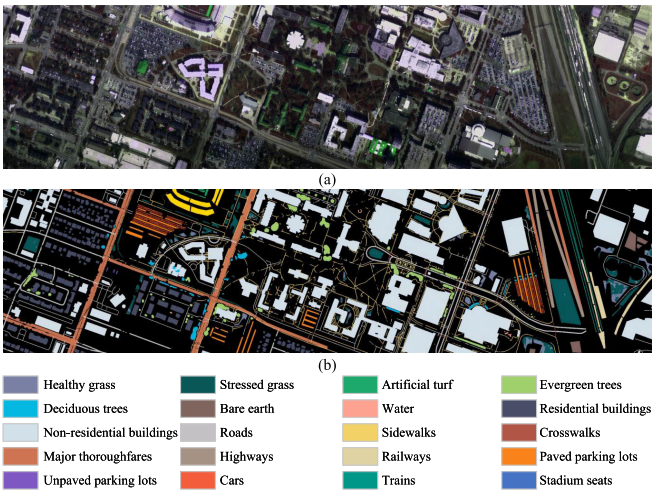


Fig. 6. HU18 dataset. (a) False-color image. (b) Ground truth.

4) Houston University 2018: The HU18 dataset, captured by CASI-1500 and provided by the NCALM for the data fusion contest of the GRSS in 2018, covers over 4 km^2. It includes 601 $\times$ 2384 pixels, 48 channels that range from 0.38 to 1.05 μm, and 20 land-cover categories in the ground truth. Fig. 6(a) and (b) shows its false-color image and ground-truth map, respectively.

B. Implementation Details

We employ ResNet [49] as the online encoder for pretraining. The encoder comprises a convolutional layer with a kernel size of 3 $\times$ 3 $\times$ 128, followed by two ResNet BasicBlocks, each containing two filters of size 3 $\times$ 3 $\times$ 128. Features from the convolutional layers are processed through an adaptive average pooling layer, producing 128-D features. The online embeddings act as anchors, while the key embeddings interact with dictionary prototypes to retrieve positive/negative examples for contrastive learning. The key encoder shares the same architecture and updates via EMA with a large momentum (e.g., $m = 0.99$ in all experiments).

Encoder pretraining runs for 200 epochs with a batch size of 32. The optimizer used is layerwise adaptive rate scaling (LARS), with a momentum of 0.99 and weight decay of 1.5×10^{-6} for updating network parameters. Dictionary updates are performed by the stochastic gradient descent (SGD) optimizer with a learning rate of 0.02, a momentum value of 0.9, and a temperature parameter τ set to 0.08 for dictionary updates and loss computation. All available samples in a dataset are used for model pretraining.

To evaluate the learned features from pre-training, a linear classifier consisting of one initialized fully connected layer (FC) is employed for linear probing. For each class in the IP, BO, HR-D, and HU18 datasets, 3% of the samples are selected for classifier training, with the remaining samples for testing. Evaluation metrics included overall accuracy (OA), average accuracy (AA), and the Kappa statistic (κ). We provide OA to assess the results of ablation experiments and parameter analysis, with all results representing the mean of ten runs. All experiments are implemented with a software environment of Python 3.8.3, PyTorch 1.7.0, and CUDA 11.0. The experiments are executed on a workstation with Ubuntu 20.04, an Intel[1] Core[2] i9-10900K CPU, and an NVIDIA GeForce RTX3090 GPU.

TABLE I
LINEAR EVALUATION RESULTS OF HIBICO WITH SHALLOW DICTIONARY CONSTRUCTED BEHIND DIFFERENT CONVOLUTIONAL LAYER s

Dictionary Location	s = 1	s = 3	s = 5	CaCo [23]
IP	**92.88**	81.71	54.80	84.26
BO	**98.89**	92.67	90.54	96.91
HR-D	**95.51**	93.77	93.23	93.95
HU18	**85.99**	81.14	80.41	83.63

TABLE II
ABLATION STUDIES ON TWO COMPONENTS IN HIBICO UNDER LINEAR PROBING ACCURACY

HiCo	BiCo	IP	BO	HR-D	HU18
		84.26	96.91	93.95	83.63
✓		85.64	93.62	94.49	84.39
	✓	90.33	95.87	95.31	85.18
✓	✓	**92.88**	**98.89**	**95.51**	**85.99**

C. Ablation Study

In HiBiCo, two main contributions are designed to facilitate discriminative HSI feature representation in SSL, including HiCo for fine-grained semantic perception and bidirectional contrastive loss to alleviate over-uniform distribution. This section will discuss several key influencing factors of them through ablation studies.

1) Where to Apply Shallow Contrastive Learning: We conducted a series of experiments to verify the influence of constructing a shallow dictionary behind the sth convolutional layers on model performance, as shown in Table I. CaCo represents the base model without any modifications. Across four datasets, the results consistently demonstrate that HiCo improves HSI representation, and that the model benefits more from a contrastive learning location further forward. Capturing low-level semantic information through an early dictionary, along with a collaborative deep dictionary that achieves image-level contrastive supervision, can help the model better understand fine-grained image semantics. In addition, separating the features early in the network and then aggregating them through subsequent convolutional layers can facilitate the exploration of discriminative features in heterogeneous hyperspectral data.

2) Effectiveness of Both Contributions: We then conduct the ablation on two main contributions and explore how they improve the base model CaCo [23]. Table II shows the ablation results, where HiCo represents the situation of HiCo, and BiCo represents the situation of bidirectional contrast. The first row corresponds to the situation where no one component is present (i.e., the base model CaCo [23]), and the last row is

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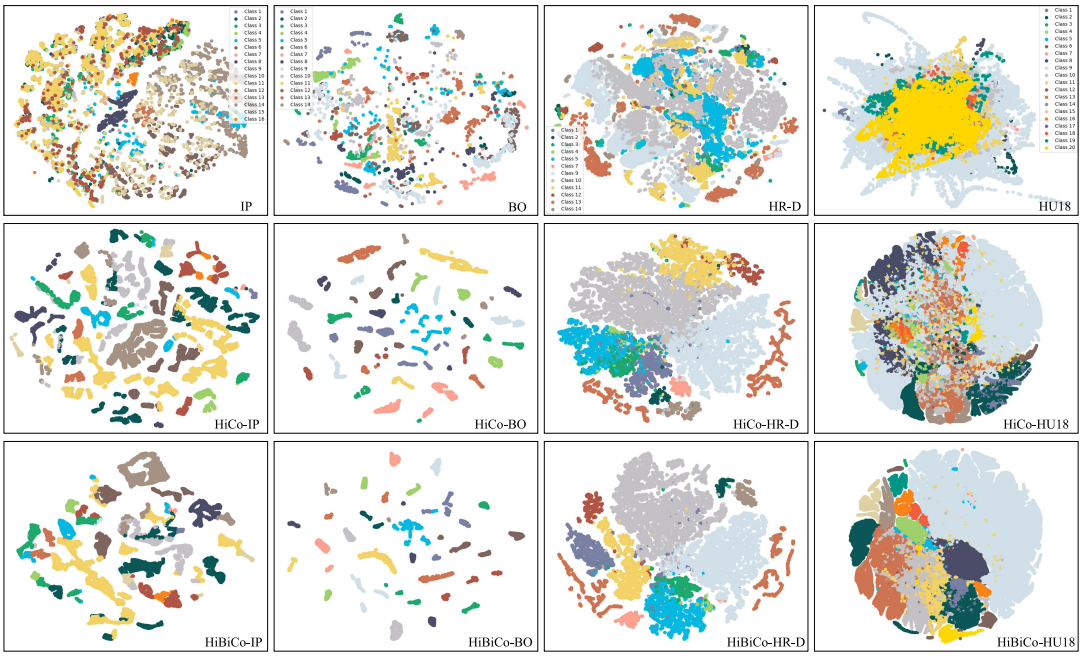


Fig. 7. Visualization of features from original images and those extracted by the proposed HiCo and HiBiCo methods across IP, BO, HR-D, and HU18 datasets.

the contrary. Both HiCo and BiCo improve performance on the IP dataset, showing that the shallow dictionary captures local spectral differences in features, while the reverse contrastive loss helps distinguish similar vegetation types. Due to spectral uncertainty, individual components perform slightly worse on the BO dataset, but benefit remarkably from their cooperation. On all four datasets, the combination of HiCo and BiCo improves the baseline model, proving their synergy in capturing diversity within classes and enhancing distinctions between classes. On the HR-D dataset, the improvement from HiCo and BiCo is less noticeable, possibly due to dictionary size limitations, as later experiments confirmed. Either the individual contribution or their corporation works on the HU18 dataset, further demonstrating its capability to handle challenging datasets.

3) Visualizations of t-SNE Feature Distribution: In this section, we employ t-SNE visualization on the features output from the online encoder across four datasets, as illustrated in Fig. 7. The first row demonstrates the original distribution of images in the feature space. The second row represents features learned from the HiCo method, which incorporates hierarchical contrastive supervision. It can be observed that features learned by HiCo exhibit a high degree of interclass separation even without parameter optimization in the classifier. However, HiCo demonstrates suboptimal intraclass clustering when processing data with lower resolution, such as IP. On the other hand, when dealing with multiclass or class-imbalanced data, the model shows limitations in handling hard negative samples, manifesting as class mixing issues, which are particularly evident in the HU18 dataset. Integrated bidirectional contrastive learning, HiBiCo further improves this. HiBiCo more effectively mitigates over-uniformity and preserves intraclass diversity, as demonstrated on the IP and HU18 datasets. At the same time, it maintains clearer clustering structures and more natural sample distributions, as seen on BO and HR-D.

TABLE III

EFFECT OF THE WEIGHT λ ON THE REVERSE CONTRASTIVE LOSS OF BICO UNDER LINEAR PROBING ACCURACY

Weight	λ=0.05	λ=0.1	λ=0.15	λ=0.2	λ=0.25	λ=0.3
IP	88.28	**92.88**	87.01	81.45	63.78	64.67
BO	**98.89**	98.10	98.03	97.30	95.68	92.22
HR-D	95.23	**95.51**	94.56	90.05	85.14	80.46
HU18	84.58	**85.99**	84.92	84.92	83.83	83.68

D. Parameter Analysis

1) Weight λ on the Reverse Contrastive Loss of BiCo: Table III shows the effect of the parameter λ in BiCo on classification performance. For a small λ, reverse contrastive learning contributes less, and the model may still face over-uniformity, leading to lower accuracy (e.g., IP: 88.28). A moderate λ (e.g., 0.10) effectively reduces over-uniformity without disrupting the alignment of positive samples by InfoNCE. However, when λ is too large, as observed in the IP dataset, accuracy drops sharply, cause the model may over-aggregate features indiscriminately and lose semantic information. Datasets with strong interclass separability, such as BO, tolerate a wider range of λ, while those with weaker separability, like IP, require careful λ selection to avoid over-separating negative samples.

2) Temperature τ in Loss Function: We investigate how the temperature τ in the overall loss function affects model performance as τ varies from 0.01 to 0.5 with quantitative results shown in Table IV. Generally, the temperature parameter τ determines the feature distribution [27], [29]. Due

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TABLE IV
EFFECT OF TEMPERATURE τ IN LOSS FUNCTION UNDER LINEAR PROBING ACCURACY

Temperature		$\tau = 0.01$	$\tau = 0.05$	$\tau = 0.08$	$\tau = 0.2$	$\tau = 0.5$
HiBiCo	IP	87.86	89.16	**92.88**	90.28	89.45
	BO	96.45	96.71	**98.89**	97.47	96.13
	HR-D	94.66	94.26	**95.51**	95.14	94.32
	HU18	81.66	85.46	**85.99**	85.11	81.69
CaCo [23]	IP	69.21	84.00	**84.26**	83.77	72.67
	BO	93.26	95.42	**96.91**	96.53	94.80
	HR-D	90.33	92.90	**93.95**	90.66	86.17
	HU18	76.72	83.01	**83.63**	82.13	74.13

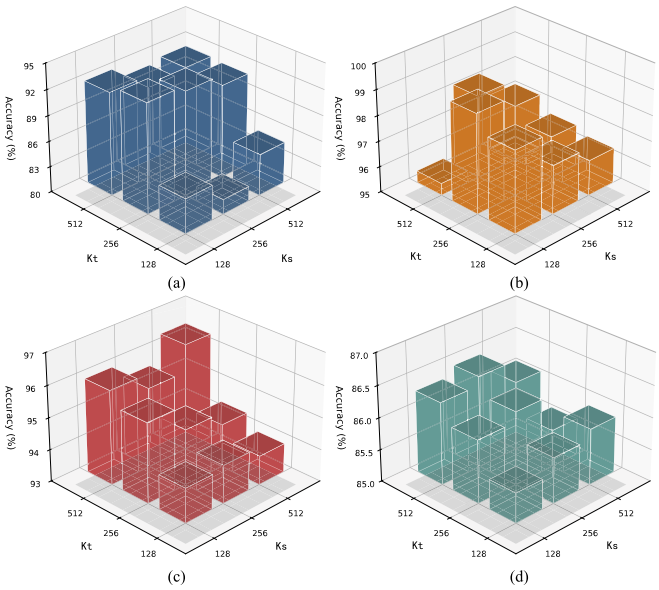


Fig. 8. Effect of prototype numbers K_s and K_t of dictionaries under linear probing accuracy for four datasets. (a) IP dataset. (b) BO dataset. (c) HR-D dataset. (d) HU18 dataset.

to the unidirectional contrast in CaCo [23], the model is highly sensitive to the temperature. While a small τ prevents model collapse, it may lead to uniform feature distribution. Whereas a large τ has the opposite effect. By introducing bidirectional information flow, HiBiCo achieves more stable decision-making for ambiguous negative samples. It is noteworthy that our HiBiCo enhances the model's robustness to temperature compared with the base model CaCo [23]. Even with a relatively larger temperature, such as 0.5, the model no longer rapidly falls into discriminability deprivation.

3) Number of Prototypes K_s and K_t in Dictionaries: We investigate the impact of the number of prototypes in the shallow dictionary (K_s) and the terminal dictionary (K_t), varying from 128 to 512, as shown in Fig. 8. Theoretically, K_s controls the capacity of the shallow layer to capture low-level features, where a smaller K_s is often sufficient to avoid redundancy and over-uniformity, especially for data with high spectral variability. In contrast, K_t governs the deep layer's ability to model high level, discriminative semantics, where a larger K_t is preferred to capture richer representations, particularly for datasets with more abundant neighborhood relationships. The choice of K_s and K_t should take both factors into account

TABLE V
EFFECT OF FEATURE DIMENSIONS d UNDER LINEAR PROBING ACCURACY

Dimension	d=32	d=64	d=128	d=256	d=512	d=1024
IP	90.34	90.62	**92.88**	89.90	86.85	87.88
BO	95.84	98.01	**98.89**	97.45	95.84	95.51
HR-D	93.39	93.63	**95.51**	94.15	94.62	95.05
HU18	85.38	85.89	**85.99**	84.83	84.77	85.84

TABLE VI
EFFECT OF BATCH SIZE B UNDER LINEAR PROBING ACCURACY

Batch Size	B=16	B=32	B=64	B=128	B=256	B=512
IP	87.98	**92.88**	88.47	86.89	84.61	77.53
BO	96.66	**98.89**	93.01	90.85	89.79	91.44
HR-D	90.26	**95.51**	94.85	94.71	94.20	94.72
HU18	81.67	85.99	87.09	**87.25**	86.83	86.33

simultaneously. Spectral uncertainty datasets like IP and BO require smaller K_s but moderate K_t to balance diversity and avoid over-uniformity, while semantically diverse datasets like HR-D and HU18 can leverage larger K_s and K_t to enhance feature representation. For instance, HR-D achieves optimal performance when both K_s and K_t are set to 512, whereas it provides excessive semantic representation capacity to BO and IP, who performs best with $K_s = 128$ and $K_t = 256$. The balancing model capacity with the characteristics of the dataset is essential for ensuring optimal performance. Considering computational overhead, default values are set uniformly in other experiments with $K_s = 128$ and $K_t = 256$.

4) Dimensions d of Feature From the Encoder: We further explore how feature dimensions (ranging from 32 to 1024) affect model performance. Results in Table V show that accuracy drops with either too low or too high-feature dimensions. Lower dimensions restrict diversity, whereas higher dimensions introduce noisy overfitting despite finer representations. All four datasets show that the model achieves the best balance between capacity and generalization at 128 dimensions.

5) Batch Size B: We further examine the effect of batch size, testing values from 16 to 512. From Table VI, we observe that batch size interacts closely with dataset characteristics. Smaller batch sizes enable frequent dictionary updates, allowing faster adaptation to local feature changes, which is beneficial for smaller datasets (such as the changes from cornfields to grasslands on IP and from water bodies to vegetation on BO). However, for larger datasets, frequent updates can destabilize the dictionary, negatively affecting generalization. Considering computational overhead, the batch size is set to 32 throughout our experiments.

E. Comparison With Other Methods

1) Competitor Methods: We compare the proposed model HiBiCo with several state-of-the-art methods for HSI feature learning, such as a SimCLR-based self-supervised contrastive learning (SSCL) method [35], which employs image spatial augmentation to construct positive pairs for contrastive learning; ContrastNet [50], which combines representations

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TABLE VII
LINEAR PROBING RESULTS WITH 3% TRAINING SAMPLES ON THE IP DATASET. THE BEST ACCURACIES ARE MARKED IN BOLD

Class	Supervised		Self-supervised								
	SC-SS-MTr	GraphGST	PCA	SSCL	ContrastNet	BYOL-T	S3FN	CaCo	LSCoSa	SSLSM	HiBiCo
1	**100**	97.78	34.09	52.46	53.33	88.89	84.44	73.24	65.69	55.56	69.57
2	77.05	**96.10**	68.02	71.04	68.66	89.36	78.70	75.69	87.75	82.24	90.80
3	83.30	92.55	68.25	62.65	89.94	77.19	76.52	80.77	84.81	89.44	**95.72**
4	98.57	66.52	52.67	51.88	90.00	49.39	85.22	89.00	71.56	84.35	**98.69**
5	84.84	89.13	84.60	71.33	86.14	**93.25**	71.64	81.19	85.05	75.69	89.06
6	93.17	**99.01**	90.94	79.56	85.31	91.91	92.94	94.02	97.59	81.78	98.06
7	**100**	**100**	36.78	80.85	70.37	61.90	74.07	48.78	56.84	92.59	31.17
8	**100**	**100**	92.43	95.38	87.50	99.68	96.77	90.66	99.89	96.12	**100**
9	**100**	68.42	21.05	47.06	63.16	72.22	63.16	0.00	27.27	**100**	57.14
10	79.37	**90.46**	61.93	72.92	65.43	80.48	79.00	82.34	90.34	90.24	88.60
11	78.76	90.60	76.68	83.95	85.85	88.99	89.71	89.14	93.37	**97.15**	94.10
12	80.98	**94.09**	56.52	61.17	78.09	74.67	80.17	74.12	89.68	76.87	89.71
13	**99.84**	98.49	89.00	89.13	80.90	92.43	92.96	92.19	99.50	80.40	99.50
14	96.76	**99.02**	87.20	88.48	90.06	95.00	91.69	91.44	94.75	95.35	94.78
15	**97.30**	64.97	59.35	53.31	72.73	70.51	84.49	60.32	76.45	84.76	81.71
16	99.18	**100**	58.02	81.82	55.56	77.02	**100**	85.25	90.53	70.00	95.45
OA(%)	85.37	92.46	73.85	76.54	80.97	86.59	85.23	84.26	90.55	88.70	**92.88**
AA(%)	**91.82**	90.45	64.85	71.44	76.44	81.43	83.84	75.51	82.34	84.53	86.12
Kappa(%)	83.48	91.42	70.38	73.10	78.28	84.70	83.15	82.02	89.04	87.03	**91.84**

TABLE VIII
LINEAR PROBING RESULTS WITH 3% TRAINING SAMPLES ON THE BO DATASET. THE BEST ACCURACIES ARE MARKED IN BOLD

Class	Supervised		Self-supervised								
	SC-SS-MTr	GraphGST	PCA	SSCL	ContrastNet	BYOL-T	S3FN	CaCo	LSCoSa	SSLSM	HiBiCo
1	85.17	**100**	94.26	92.29	87.02	95.58	**100**	96.68	96.32	99.18	97.04
2	95.74	96.94	78.05	62.62	87.76	98.62	**100**	93.01	**100**	92.31	**100**
3	86.72	**100**	88.20	97.93	94.65	98.87	97.94	**100**	**100**	**100**	99.79
4	99.09	**100**	87.69	86.08	98.56	97.63	98.56	98.10	**100**	**100**	**100**
5	86.87	89.65	78.98	81.04	84.29	88.21	89.66	93.10	93.05	93.80	**96.47**
6	85.92	95.79	81.53	82.66	98.85	**99.19**	80.84	90.65	**100**	89.26	97.36
7	99.92	**100**	94.30	95.47	99.20	**100**	99.60	**100**	**100**	**100**	**100**
8	91.99	**100**	94.44	87.78	98.98	99.95	**100**	99.98	**100**	98.36	98.99
9	**100**	**100**	95.72	97.18	**100**	97.61	99.02	99.01	**100**	**100**	99.84
10	**100**	98.34	55.54	89.76	**100**	97.25	**100**	**100**	**100**	**100**	**100**
11	95.20	**100**	89.19	90.36	95.95	99.38	97.64	99.00	**100**	97.82	**100**
12	**100**	81.25	73.18	87.30	98.86	94.47	94.89	88.75	**100**	97.55	**100**
13	**100**	98.46	87.97	83.37	**100**	99.31	99.62	99.31	**100**	99.59	**100**
14	90.23	**100**	85.11	89.01	80.43	90.48	82.61	90.48	90.48	78.82	90.48
OA(%)	94.05	97.40	86.55	88.75	95.49	97.18	96.13	96.91	98.85	97.20	**98.89**
AA(%)	94.06	97.17	84.58	87.35	94.61	96.90	95.74	96.29	98.56	96.19	**98.57**
Kappa(%)	93.56	97.18	85.33	87.80	95.12	96.94	95.80	96.65	98.76	96.96	**98.80**

obtained from two generative networks adversarial autoencoder (AAE) and variational autoencoder (VAE) as positives and integrates prototypical contrastive loss; BYOL-T [51], a BYOL-based SSCL method using Transformer backbone; the self-supervised spatial–spectral feature learning network (S3FN) [52], which performs the contrastive loss to align spectral–spatial representations from high-level semantic perspectives; the CaCo model [23], sharing the same backbone with HiBiCo; the LSCoSa method [24], which is based on the sparse dictionary training and prototypical contrast; and SSLSM [53], a self-supervised feature learning method based on masked spectral reconstruction. We also choose two supervised methods, containing a hybrid model SC-SS-MTr [54], which incorporated labeled data, mask reconstruction, and contrastive representation learning; and a graph generative structure-aware Transformer-based method GraphGST [55], incorporated with self-supervised cross-view contrastive learning. In addition, we present a baseline accuracy from a radial basis function-based support vector machine (RBF-SVM) with the first 15 components from principal component analysis (PCA) as features for HSIC performance evaluation.

Tables VII–X report the experimental results of all selected comparative methods on the IP, BO, HR-D, and HU18 datasets. All results are averaged over ten runs, and the quantification metrics include per class accuracy, OA, AA, and kappa. For all comparative methods, the image input size is fixed at 15 × 15. For self-supervised pre-trained models, we utilize the same initialized FC layers for linear probing. For

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TABLE IX
LINEAR PROBING RESULTS WITH 3% TRAINING SAMPLES ON THE HR-D DATASET. THE BEST ACCURACIES ARE MARKED IN BOLD

Class	Supervised		Self-supervised								
	SC-SS-MTr	GraphGST	PCA	SSCL	ContrastNet	BYOL-T	S3FN	CaCo	LSCoSa	SSLSM	HiBiCo
1	88.37	82.60	74.33	75.75	83.43	80.41	73.28	86.56	87.25	84.97	**89.09**
2	**100**	94.44	68.45	84.09	70.86	96.62	98.99	98.41	97.27	84.34	99.98
3	82.27	82.55	69.12	54.60	53.74	74.66	**89.09**	80.46	85.97	65.10	85.68
4	**92.00**	71.23	14.40	58.16	33.77	52.36	70.55	82.16	81.03	57.53	78.66
5	90.51	**93.29**	84.69	73.99	86.15	82.76	84.14	91.64	91.59	91.20	90.65
6	**100**	**100**	88.29	94.25	91.03	97.18	66.29	99.04	**100**	88.00	98.55
7	92.14	94.19	89.74	90.00	89.45	91.32	91.58	95.66	96.19	94.76	**96.40**
8	90.99	**97.65**	88.33	87.84	89.18	89.38	95.96	95.45	96.51	97.27	96.80
9	96.60	**97.30**	77.85	78.88	90.45	80.20	93.18	89.54	95.85	93.24	96.30
10	**99.10**	90.36	83.33	85.42	94.13	92.47	93.50	90.58	93.74	84.07	95.03
11	99.74	**100**	99.47	96.57	99.81	**100**	**100**	99.83	**100**	**100**	**100**
12	**100**	98.96	94.60	82.00	97.51	99.87	94.82	99.28	99.35	98.96	99.52
OA(%)	92.59	94.80	86.10	85.03	88.21	88.62	91.48	93.95	95.29	93.32	**95.51**
AA(%)	**94.31**	91.88	77.72	80.13	81.63	86.44	87.62	92.38	93.69	86.62	93.89
Kappa(%)	90.82	93.54	82.93	81.42	85.42	85.71	89.41	92.49	94.14	91.67	**94.42**

TABLE X
LINEAR PROBING RESULTS WITH 3% TRAINING SAMPLES ON THE HU18 DATASET. THE BEST ACCURACIES ARE MARKED IN BOLD

Class	Supervised		Self-supervised								
	SC-SS-MTr	GraphGST	PCA	SSCL	ContrastNet	BYOL-T	S3FN	CaCo	LSCoSa	SSLSM	HiBiCo
1	86.98	**90.59**	61.73	70.45	62.46	75.90	58.43	78.27	72.71	69.98	78.41
2	79.39	86.23	80.85	80.89	88.55	84.48	85.70	88.93	85.95	79.50	**89.59**
3	**100**	99.70	60.94	92.15	47.14	98.08	97.49	99.24	**100**	99.70	97.84
4	**97.52**	93.90	84.38	83.10	93.93	89.19	91.67	92.32	83.91	84.14	92.41
5	**87.53**	75.54	57.46	54.51	45.83	56.90	22.80	65.94	54.00	61.68	76.47
6	**100**	97.19	71.32	94.18	71.24	75.03	36.99	98.20	96.30	96.96	97.12
7	84.86	**99.61**	48.33	83.37	62.40	82.19	97.65	95.12	78.59	85.66	90.57
8	**94.39**	88.31	74.56	75.95	69.75	81.08	61.19	86.44	79.87	88.14	88.32
9	84.27	96.50	91.71	85.18	90.05	92.99	**97.64**	92.33	90.78	96.33	94.84
10	53.47	63.27	51.80	46.76	45.77	52.37	**87.55**	58.99	50.23	72.83	64.13
11	46.10	40.91	48.16	40.93	42.37	44.93	**64.12**	50.29	47.31	51.60	57.06
12	**78.03**	11.35	5.48	16.03	0.00	0.94	9.23	2.64	0.14	1.77	7.54
13	72.01	78.85	59.23	60.30	56.15	65.43	**79.23**	73.68	68.57	65.35	75.17
14	**99.17**	73.23	60.57	75.42	73.35	85.65	87.16	90.93	81.47	90.56	88.83
15	**97.75**	97.16	70.87	82.68	92.61	82.19	89.99	93.84	79.52	94.56	93.10
16	**96.69**	84.49	78.81	67.39	63.56	44.17	64.42	84.71	84.39	85.89	88.63
17	**100**	**100**	18.85	53.16	1.41	35.92	92.24	95.97	45.65	70.40	95.05
18	**95.12**	85.20	61.94	55.64	23.32	56.85	75.84	84.39	78.63	54.34	88.33
19	**96.56**	88.41	77.97	77.57	47.47	90.85	80.25	95.31	88.87	83.33	96.27
20	**100**	99.06	87.04	94.94	62.82	95.59	95.04	97.49	98.85	94.30	97.39
OA(%)	80.02	85.23	76.88	74.28	74.02	79.17	83.18	83.63	80.04	84.02	**85.99**
AA(%)	**87.49**	82.48	62.60	69.53	57.01	69.54	73.73	81.25	73.29	72.78	82.85
Kappa(%)	74.86	80.72	69.97	66.57	66.20	72.93	80.32	78.55	73.73	79.14	**81.73**

supervised methods, we replace the classifier with the same initialized FC. Other parameters involved in the models are set according to default values from the corresponding references.

2) Experimental Results: From the experimental results in Tables VII–X, it can be observed that all the models enhance data representation compared to the PCA baseline, except for SSCL and ContrastNet, which show slightly inferior performance on HR-D and HU18. SSCL [35] primarily employs spatial data augmentation, which has been proved not suitable for hyperspectral data with vague spatial semantics. ContrastNet [50] utilizes two generative encoders VAE and AAE as a positive pair creator, instead of a simple augmentation, resulting in stable improvements across most testing datasets, but cannot match the results by BYOL-T. BYOL [56] is famous for the independence on negative samples, which shows a great combination for the challenging design of data augmentation in HSI representation. BYOL-T demonstrates much better performance on four datasets compared with ContrastNet and SSCL, two SimCLR-based methods relying on specific image augmentation. At the same time, due to special designs such as EMA and stop-gradient for feature consistency, BYOL-T achieves comparable results to CaCo [23], a method that also places great emphasis on feature uniformity. With end-to-end prototypical dictionary updating, CaCo [23] performs remarkably well on the HR-D and HU18 datasets, where the supervision of hard samples counts much.

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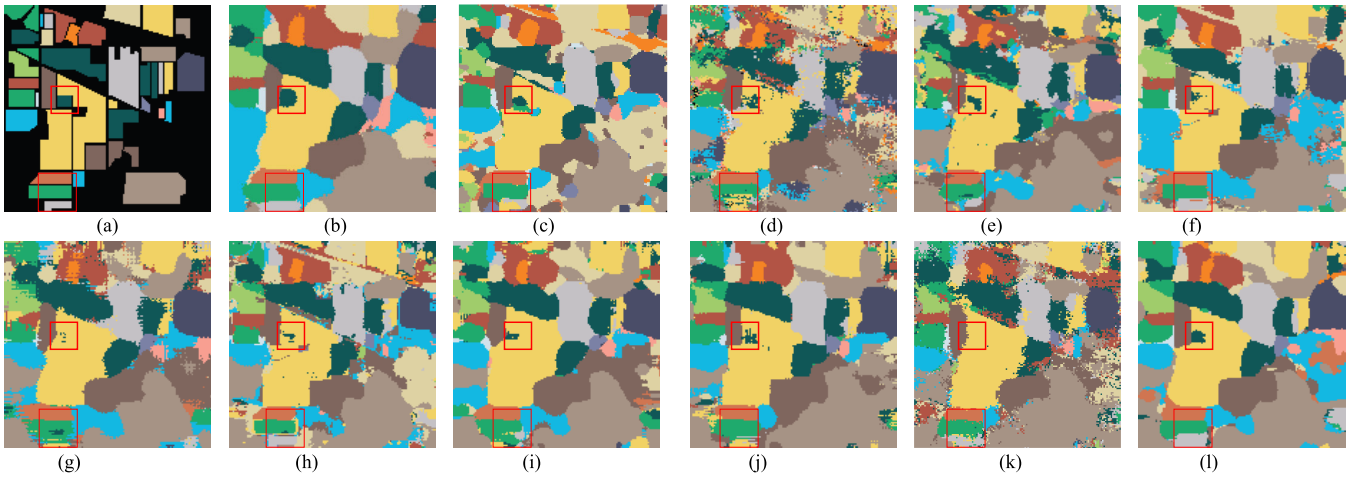


Fig. 9. Classification maps of the IP dataset produced by (a) ground truth, (b) SC-SS-MTr [54], (c) GraphGST [55], (d) PCA, (e) SSCL [35], (f) ContrastNet [50], (g) BYOL-T [51], (h) S3FN [52], (i) CaCo [23], (j) LSCoSa [24], (k) SSLSM [53], and (l) HiBiCo.

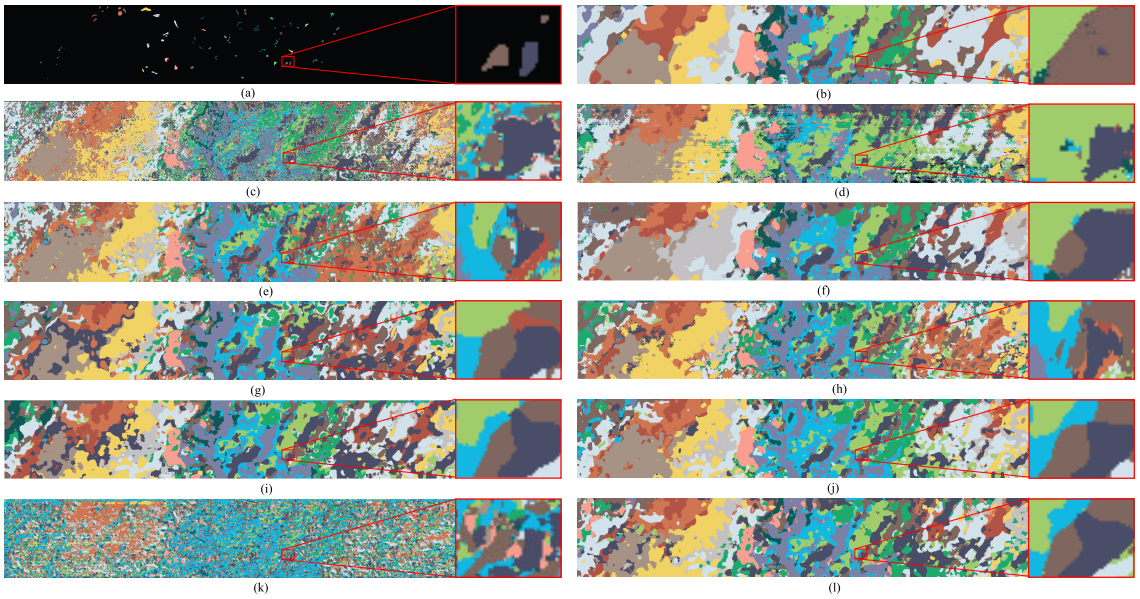


Fig. 10. Classification maps of the BO dataset produced by (a) ground truth, (b) SC-SS-MTr [54], (c) GraphGST [55], (d) PCA, (e) SSCL [35], (f) ContrastNet [50], (g) BYOL-T [51], (h) S3FN [52], (i) CaCo [23], (j) LSCoSa [24], (k) SSLSM [53], and (l) HiBiCo.

Leveraging specific spectral and spatial features and aligning them with contrastive loss, S3FN [52] exhibits superior performance on the HR-D and HU18 datasets compared with BYOL-T [51], but fails to maintain consistent advantages on the other datasets. SSLSM [53] uses masked spectral reconstruction for pretraining, achieving accuracy comparable to CaCo [23]. However, this spectral-focused approach weakens spatial awareness, leading to limited neighborhood preservation under linear probing. Excessive representation uniformity may hurt the preservation of the data manifold. LSCoSa [24] improves CaCo [23] through sparse sampling and multi-positive contrast, resulting in significant accuracy improvements across all datasets except the semantically rich dataset HU18. Supervised methods like SC-SS-MTr [54] and GraphGST [55] effectively address imbalanced classes through label supervision, performing excellently in classes with limited samples (such as the seventh and ninth classes of the IP dataset, and the 7th and 19th classes of the HU18 dataset). SC-SS-MTr [54] thus achieving the best AA on the IP, HR-D and HU18 datasets. However, the label supervision cannot capture the intrinsic data essence well, SC-SS-MTr obtains a limited OA and Kappa in all experimental datasets. GraphGST [55] designs an absolute positional encoding to capture pixel positional sequences and integrates a generative self-supervised mechanism for enhanced representation learning. Leveraging its powerful capability to uncover latent relationships in data, GraphGST achieves outstanding classification accuracy on four experimental datasets, while also attaining the best accuracy for most classes in the IP, BO, and HR-D datasets.

Self-supervision at network terminal with learnable dictionary could obscure specific semantics in hyperfine spectra, and the over-uniformity may shatter the data manifold, too. HiBiCo improves CaCo by hierarchical dictionary and bidirectional contrast, performing an 8.62%, 1.98%, 1.56%, and 2.36% OA improvement on the testing dataset, respectively. In terms of

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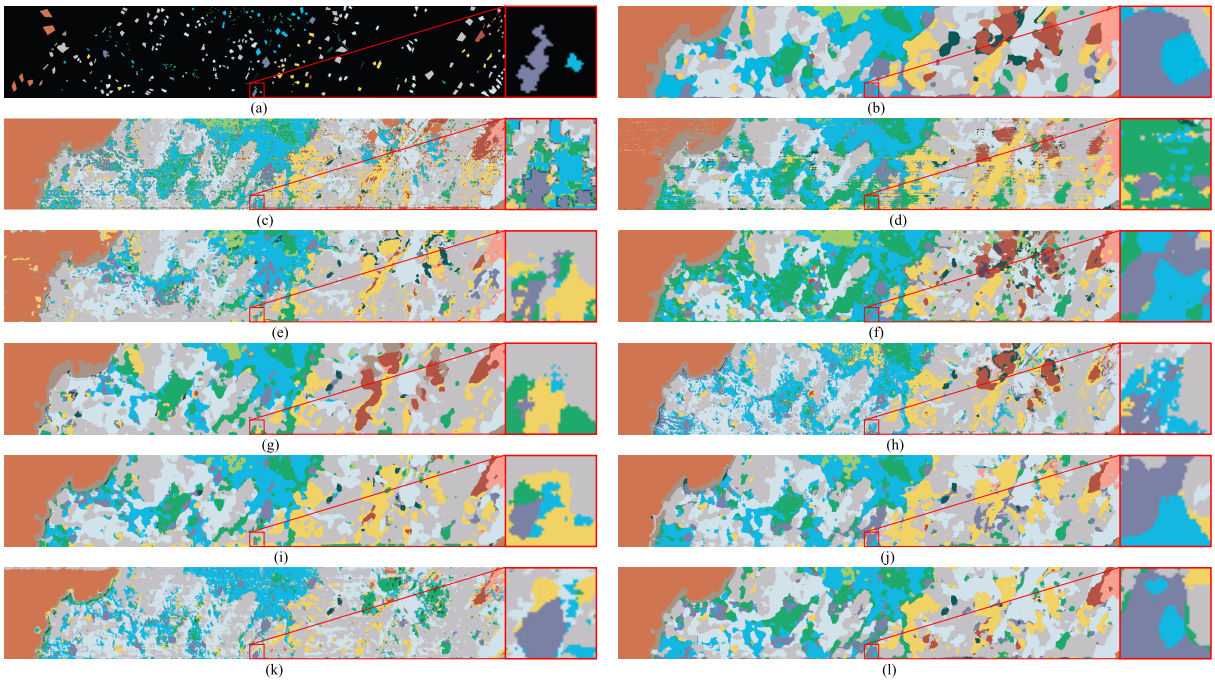


Fig. 11. Classification maps of the HR-D dataset produced by (a) ground truth, (b) SC-SS-MTr [54], (c) GraphGST [55], (d) PCA, (e) SSCL [35], (f) ContrastNet [50], (g) BYOL-T [51], (h) S3FN [52], (i) CaCo [23], (j) LSCoSa [24], (k) SSLSM [53], and (l) HiBiCo.

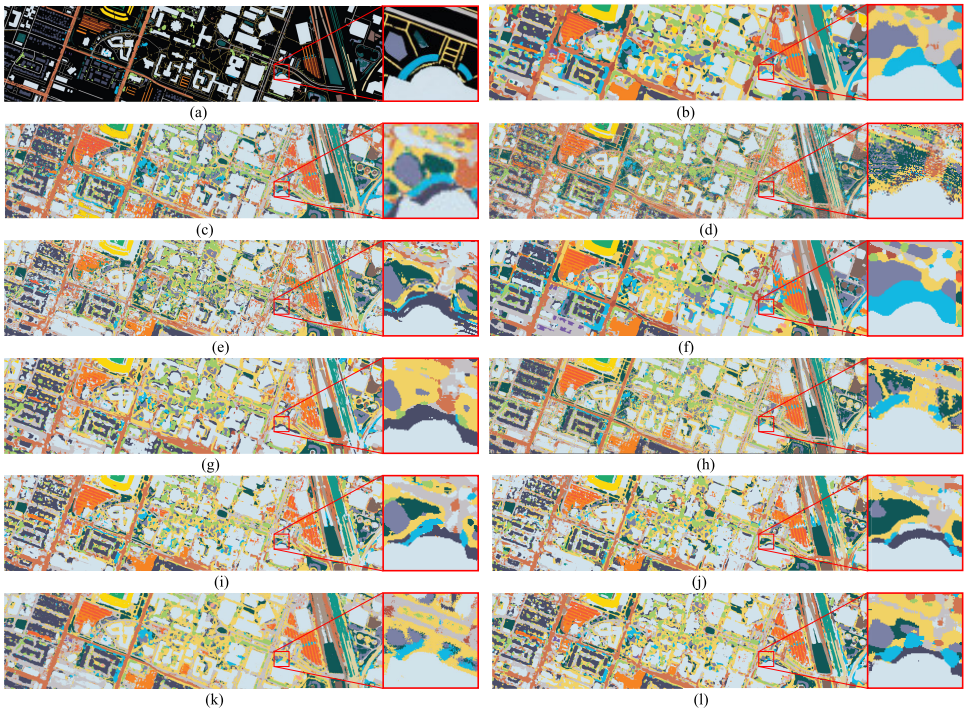


Fig. 12. Classification maps of the HU18 dataset produced by (a) ground truth, (b) SC-SS-MTr [54], (c) GraphGST [55], (d) PCA, (e) SSCL [35], (f) ContrastNet [50], (g) BYOL-T [51], (h) S3FN [52], (i) CaCo [23], (j) LSCoSa [24], (k) SSLSM [53], and (l) HiBiCo.

self-supervised approaches, HiBiCo also achieves the best results in most categories across the four datasets. The IP dataset features typical class imbalance and spectral uncertainty. Learning highly discriminative representations from images is particularly challenging when the training set is very few. The ninth class in the IP dataset is quite unique, with just 20 samples representing a very minor land cover. Aside from labeled supervision, self-supervised methods generally struggle with such data. CaCo even completely overlooks this class information with only 3% training set. Although HiBiCo does not achieve high classification accuracy for this class, it effectively considers the class imbalance and improves CaCo further (0% $\rightarrow$ 57.14% OA accuracy).

Visualizations of the classification maps for the four datasets are presented in Figs. 9–12. It can be observed that HiBiCo exhibits more consistent regional categories and more precise

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TABLE XI
COMPUTATIONAL COST OF VARIOUS METHODS ON IP DATASET

Methods	Supervised		Self-supervised								
	SC-SS-MTr	GraphGST	PCA	SSCL	ContrastNet	BYOL-T	S3FN	CaCo	LSCoSa	SSLSM	HiBiCo
Params(M)	1.4587	1.3324	/	2.3144	1.9511	25.2838	11.2248	**0.8878**	**0.8878**	33.5436	**0.8878**
FLOPs(G)	201.4180	**0.1491**	/	1.7017	154.9893	1.9755	0.2783	8.8543	8.8543	96.7740	3.3178
Running time(s)	3.3056	6.1624	**2.5306**	27.9571	26.8269	45.4708	25.4073	10.4121	11.9901	26.0183	8.7461

boundaries in overall classification performance compared with other methods. Taking the IP dataset as an example, although HiBiCo suffers from misclassification of some small-area features, it offers more precise boundary positioning, such as the multiclass mixed areas highlighted in red. Furthermore, in the context of fine-grained object recognition, it presents a challenge to achieve a balance between neighborhood smoothness and boundary localization. For instance, when applied to the BO dataset, the SSLSM [53] method demonstrates a terrible neighborhood distribution. This occurrence is attributable to its masked spectral reconstruction, which prioritizes spectral perception at the expense of spatial supervision, consequently leading to a notable disadvantage when the encoder is frozen for linear probing. Our proposed approach, HiBiCo, achieves an improved balance between neighborhood smoothness and boundary localization.

3) Computational Cost Results: This section presents the time consumption of all HSIC methods in Table XI on the IP dataset. For supervised methods, although SC-SS-MTr shows suboptimal classification performance, it possesses high execution efficiency second only to PCA. With the lowest computational complexity, GraphGST strikes a balance between efficiency and performance. Self-supervised methods generally need to occupy more execution time than supervised approaches due to their complexity, but offer scalability in label-scarce scenarios. Although the HiBiCo method does not have an advantage over GraphGST, SSCL, and BYOL-T in terms of computational complexity, it has a strong advantage in terms of the comprehensive consideration of classification accuracy, execution efficiency, and model complexity.

4) Linear Probing on Small Training Samples: SSL derives valuable information from data essence rather than the manually annotated labels to mitigate overfitting, and the accuracy of linear probing on small training samples can effectively reflect the feature generalization ability of the model. By selecting training samples for each class ranging from 1% to 15%, we compare the feature learning performance of the proposed HiBiCo model with all the comparative methods under linear probing. From the results in Fig. 13, it can be observed that HiBiCo achieves outstanding accuracy improvements over the base model CaCo [23] across all proportions of training sets in four datasets. Meanwhile, HiBiCo achieves state-of-the-art performance among self-supervised approaches on IP, HR-D, and HU18, being outperformed only by SSLSM [53] at limited training ratios, a marginal advantage that comes at 10× higher computational cost. Notably, HiBiCo demonstrates comparable accuracy to the supervised GraphGST [55] baseline in most scenarios. HiBiCo under 1% training samples on BO is inferior to BYOL-T [51] and

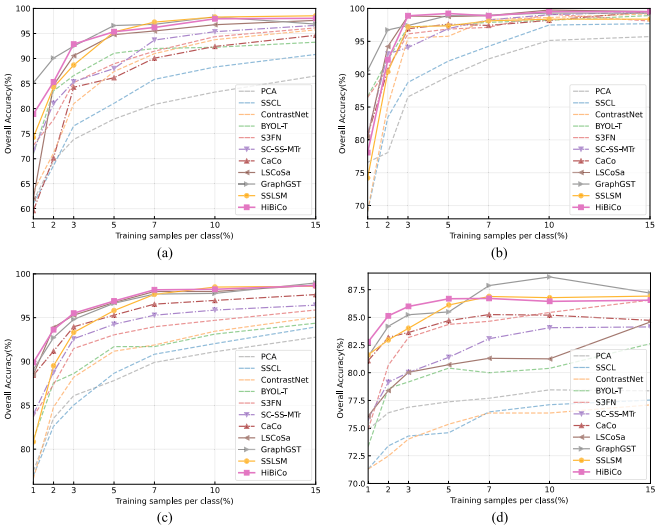


Fig. 13. Linear probing results of all comparative methods under different training samples across four datasets. (a) IP dataset. (b) BO dataset. (c) HR-D dataset. (d) HU18 dataset.

S3FN [52], who are started from PCA preprocess, seem to expand initial performance but limit the upper. HiBiCo is slightly inferior to the LSCoSa at 1% and 2% training sets in BO. This could probably be attributed to the lower level of category mixture of the BO dataset compared to the other datasets, as shown in Fig. 7. HiBiCo involves hierarchical contrastive supervision to ensure representation consistency, which may be less critical on less challenging datasets. In contrast, LSCoSa [24] mitigates this by utilizing sparse dictionary sampling. In general, HiBiCo uses a hierarchical approach to separate features and improve intraclass clustering through bidirectional contrast, making it especially beneficial for datasets with a hyperfine spectrum.

V. CONCLUSION

This article presents a HiBiCo method for network pretraining. Due to severe spectral uncertainty in HSI, HiBiCo directly learns both positives and negatives through end-to-end dictionary training. Meanwhile, a dual-dictionary based deep supervision is built to enhance the utilization of shallow details in contrastive learning, by which the model could better capture specific semantics in hyperfine spectra. Furthermore, we introduce reverse contrastive learning under negative sample dominance to address excessively uniform representations. Experiments demonstrate that the HiBiCo outperforms the state-of-the-art self-supervised methods on the HSIC task.

Contrastive learning with learnable prototypes requires fixity of lower temperature to prevent model collapse, but lacks

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dynamic cognition and penalty for hard negatives. Therefore, our future work will attempt to explore an alignment-adaptive pretraining model.

REFERENCES

[1] Z. Sun, D. Wu, and Y. Lv, "Optical properties of snow surfaces: Multiangular photometric and polarimetric hyperspectral measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4301516.

[2] C. Zhou, Z. He, A. Lou, and A. Plaza, "RGB-to-HSV: A frequency-spectrum unfolding network for spectral super-resolution of RGB videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609318.

[3] F. A. Kruse, J. W. Boardman, and J. F. Huntington, "Comparison of airborne hyperspectral data and EO-1 hyperion for mineral mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1388–1400, Jun. 2003.

[4] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[5] Y. Su et al., "DAAN: A deep autoencoder-based augmented network for blind multilinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512715.

[6] C. Zhou, Z. He, G. Lai, and A. Plaza, "A selective semantic transformer for spectral super-resolution of multispectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 7436–7450, 2025.

[7] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[8] Y. Su et al., "ACGT-Net: Adaptive cuckoo refinement-based graph transfer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521314.

[9] C. Zhou, Z. He, L. Zou, Y. Li, and A. Plaza, "HSACT: A hierarchical semantic-aware CNN-transformer for remote sensing image spectral super-resolution," *Neurocomputing*, vol. 636, Jul. 2025, Art. no. 129990.

[10] Y. Zhang, P. Duan, L. Liang, X. Kang, J. Li, and A. Plaza, "PFS3F: Probabilistic fusion of superpixel-wise and semantic-aware structural features for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, early access, 2025, doi: 10.1109/TCSVT.2025.3556548.

[11] X. Cao, J. Yao, Z. Xu, and D. Meng, "Hyperspectral image classification with convolutional neural network and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4604–4616, Jul. 2020.

[12] X. Zhang et al., "Spectral–spatial distribution consistent network based on meta-learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520915.

[13] F. Liu, W. Gao, J. Liu, X. Tang, and L. Xiao, "Adversarial domain alignment with contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525720.

[14] W. Xie, Z. Zhang, L. Jiao, and J. Wang, "Decoupled knowledge distillation via spatial feature blurring for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8938–8955, 2024.

[15] Y. Yang et al., "Semi-supervised multiscale dynamic graph convolution network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 5, pp. 6806–6820, May 2024.

[16] B. Xi, Y. Zhang, J. Li, Y. Li, Z. Li, and J. Chanussot, "CTF-SSCL: CNN-transformer for few-shot hyperspectral image classification assisted by semisupervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532617.

[17] Y. Zhou, X. Huang, X. Yang, J. Peng, Y. Ban, and N. Jiang, "MSMT-LCL: Multi-scale spatial–spectral masked transformer with local contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534316.

[18] Y. Chen and L. Bruzzone, "Self-supervised change detection by fusing SAR and optical multi-temporal images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 3101–3104.

[19] H. Li et al., "Global and local contrastive self-supervised learning for semantic segmentation of HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618014.

[20] X. Li, D. Shi, X. Diao, and H. Xu, "SCL-MLNet: Boosting few-shot remote sensing scene classification via self-supervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5801112.

[21] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

[22] Q. Hu, X. Wang, W. Hu, and G.-J. Qi, "AdCo: Adversarial contrast for efficient learning of unsupervised representations from self-trained negative adversaries," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1074–1083.

[23] X. Wang, Y. Huang, D. Zeng, and G.-J. Qi, "CaCo: Both positive and negative samples are directly learnable via cooperative-adversarial contrastive learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10718–10730, Sep. 2023.

[24] M. Liang, J. Dong, L. Yu, X. Yu, Z. Meng, and L. Jiao, "Self-supervised learning with learnable sparse contrastive sampling for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5530713.

[25] N. Park, W. Kim, B. Heo, T. Kim, and S. Yun, "What do self-supervised vision transformers learn?" in *Proc. 11th Int. Conf. Learn. Represent.*, 2023, pp. 1–19.

[26] S. Ren, F. Wei, S. A. Z. Zhang, and H. Hu, "DeepMIM: Deep supervision for masked image modeling," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Feb. 2025, pp. 879–888.

[27] F. Wang and H. Liu, "Understanding the behaviour of contrastive loss," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2495–2504.

[28] T. Wang and P. Isola, "Understanding contrastive representation learning through alignment and uniformity on the hypersphere," in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, 2020, pp. 9929–9939.

[29] Z. Huang et al., "Model-aware contrastive learning: Towards escaping the dilemmas," in *Proc. 40th Int. Conf. Mach. Learn.*, vol. 202, 2023, pp. 13774–13790.

[30] Z. Han, J. Yang, L. Gao, Z. Zeng, B. Zhang, and J. Chanussot, "Dual-branch subpixel-guided network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5521813.

[31] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[32] H. Yang, H. Yu, K. Zheng, J. Hu, T. Tao, and Q. Zhang, "Hyperspectral image classification based on interactive transformer and CNN with multilevel feature fusion network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[33] C. Zhou, Z. He, J. Dong, Y. Li, J. Ren, and A. Plaza, "Low-rank and sparse representation meet deep unfolding: A new interpretable network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5513516.

[34] L. Jing, P. Vincent, Y. LeCun, and Y. Tian, "Understanding dimensional collapse in contrastive self-supervised learning," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–17.

[35] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5521213.

[36] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[37] L. Zhao et al., "Purified contrastive learning with global and local representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520414.

[38] B. Li, L. Fang, N. Chen, J. Kang, and J. Yue, "Enhancing hyperspectral image classification: Leveraging unsupervised information with guided group contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504317.

[39] Z. Ye et al., "Self-supervised learning with multiscale densely connected network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522615.

[40] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.

[41] L. Wang, C.-Y. Lee, Z. Tu, and S. Lazebnik, "Training deeper convolutional networks with deep supervision," 2015, *arXiv:1505.02496*.

[42] L. Zhang, X. Chen, J. Zhang, R. Dong, and K. Ma, "Contrastive deep supervision," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2022, pp. 1–19.

[43] X. Huang, M. Dong, J. Li, and X. Guo, "A 3-D-Swin transformer-based hierarchical contrastive learning method for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5411415.

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[44] S. Chang, M. Kopp, P. Ghamisi, and B. Du, "Dsfer-Net: A deep supervision and feature retrieval network for bitemporal change detection using modern Hopfield networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5631914.
[45] H. Zheng et al., "Multifeature collaborative fusion network with deep supervision for SAR ship classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5212614.
[46] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.
[47] C. Feng and I. Patras, "Adaptive soft contrastive learning," in *Proc. 26th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2022, pp. 2721–2727.
[48] M. Zheng et al., "ReSSL: Relational self-supervised learning with weak augmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 2543–2555.
[49] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[50] Z. Cao, X. Li, Y. Feng, S. Chen, C. Xia, and L. Zhao, "ContrastNet: Unsupervised feature learning by autoencoder and prototypical contrastive learning for hyperspectral imagery classification," *Neurocomputing*, vol. 460, pp. 71–83, Oct. 2021.
[51] X. Hu, T. Li, T. Zhou, Y. Liu, and Y. Peng, "Contrastive learning based on transformer for hyperspectral image classification," *Appl. Sci.*, vol. 11, no. 18, p. 8670, Sep. 2021.
[52] H. Xu, W. He, L. Zhang, and H. Zhang, "Unsupervised spectral–spatial semantic feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526714.
[53] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715.
[54] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508718.
[55] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.
[56] J.-B. Grill et al., "Bootstrap your own latent-a new approach to self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21271–21284.



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